
Named Entity Recognition in German-language Telegram channels

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Abstract

This thesis investigates the potentials and limitations of Named Entity Recognition (NER) approaches for the automated analysis of text data from the messenger service Telegram.

The thematic focus is on the mobilizations for protests against the COVID-19 measures in Germany. The rise of conspiracy theories in the context of the COVID-19 pandemic not only reinforced antisemitic and racist discrimination against certain communities, but also generated an increasingly toxic and violent attitude towards political decision-makers and scientific experts.

The past months have once again shown the importance of investigative work by critical journalists and non-governmental organizations to critically monitor the corresponding mobilization strategies on Telegram. A central challenge here is the large amounts of user-generated data from often very dynamic contexts, which are usually analyzed manually. Here, approaches from the field of natural language processing can offer a valuable addition.

Against this background, this thesis aims to evaluate the applicability of Named Entity Recognition for the investigation of the COVID-19 protest movement in German-language Telegram channels, focusing on entities representing people, places, organizations, actions, dates, and times.

This work contributes to the development of German-language Named Entity Recognition in general by applying this approach to a specific text genre and domain, and provides first insights for developing an NER tool for researchers and non-governmental organizations in the future.

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Chapter 1

Introduction

1.1 Background

The outbreak of the COVID-19 pandemic was followed by an increasing spread of misinformation and conspiracy theories. Those narratives evolved not only around the virus itself, but also about the intentions of those responsible for putting into effect measures for its containment. Those measures included the mandatory use of face masks, rapid testing, and vaccinations, at times functioning as admission requirements in the workplace or publicly accessible spaces such as restaurants and theatres. Furthermore, unprecedented restrictions were imposed on public life through several general or partial lock-downs of social and economic life, as well as temporary bans of gathering in public. These restrictions also heavily affected the possibilities to hold political demonstrations. From the beginning on, these measures were accompanied by critique and protests from various political spectres. In Germany, the so-called ‘Querdenken’ (lateral thinking) movement soon became the most prominent network organizing protests against the COVID-19 containment measures. The movement reached its preliminary climax with several large demonstrations taking place in Berlin, directed against legislative changes regarding the containment of the pandemic passed by the German parliament. In the context of these demonstrations, violent clashes

occurred between police and protesters, which often refused to act according to the official restrictions such as wearing a face mask. Furthermore, there were several incidents in which journalists were violently attacked by protesters and accused of being part of the ‘Lügenpresse’ (lying press). Furthermore, there were two incidents in which the seat of the German parliament, the Reichstag, was targeted by protesters.

On April 21st 2021, the last large authorized demonstration of the ‘Querdenken’ movement took place in Berlin. In response to several violations of official restrictions by demonstration participants, police decided to disperse the demonstration. Several protesters subsequently moved to other areas nearby where clashes between protesters and police continued to occur throughout the rest of the day.

The subsequent demonstrations announced by the ‘Querdenken’ movement to take place in Berlin were prohibited by authorities, arguing with the experience at prior demonstrations that participants would act against the official restrictions. In spite of the ban, several hundred protesters gathered in different spots in Berlin e.g. on the 21st, 28th and 29th of August 2021.

The mentioned bans of large demonstrations of the ‘Querdenken’ movement as well as the general restrictions for demonstrations and public gatherings (such as limiting the number of participants to a maximum of twenty persons), decreased the momentum of the large, centralized protests focussing on the German legislative located in Berlin. However, the protests did not stop as a result, but rather changed their character: They became more decentralized and a shift to more informal forms of gatherings and protests could be observed: In several German cities and towns, people gathered for ‘Mahnwachen’ or ‘Spaziergänge’ (walks) to protest against the government’s COVID-19 politics. Since going for a walk individually was a legal outdoor activity even under the strictest lock-down regulations, this type of action was chosen to circumvent the restrictions imposed on demonstrations. In effect, these walks (‘Spaziergänge’) often resulted in large gatherings of people moving through the streets together, often holding signs or chanting slogans

against German politicians or the COVID-19 containment measures in general.

On some occasions, these walks turned into gatherings in front of the private homes of politicians in charge for the COVID-19 measures on federal state level: In December 2021, around thirty members of the far-right group ‘Freie Sachsen’ (Free Saxonians) gathered in front of the home of Saxony’s Minister of Health, Petra Köpping, carrying torchlights. The incident received a lot of public attention, and was criticized in the media as well as by German politicians as attempts to intimidate politicians under the pretext of the protesters right to ‘freedom of speech’. Earlier that year, the same group had gathered in front of Saxony’s prime minister, Michael Kretschmer, who is one of the most prominent targets of the radical movement against the COVID-19 containment measures. In December 2021, an investigative report by German journalists revealed that members of the Telegram chat group ‘Dresden Offlinevernetzung’ had talked about concrete preparations to assassinate Kretschmer. And a few months later, by mid-April 2022, police foiled a plot to kidnap Germany’s Minister of Health Karl Lauterbach and to sabotage power facilities in order to create chaos which would ultimately allow to overthrow the government. The rise of conspiracy theories in the context of the COVID-19 pandemic has thus not ‘only’ intensified hate-speech against and the discrimination of certain communities in the form of antisemitism and racism - this process was accompanied by an increasingly toxic and potentially violent climate for politicians in Germany.

The messenger service Telegram played a key role in the mobilization of the movement. The service allows the creation of large groups and channels to disseminate information quickly to a large number of users. Unlike Facebook or Google, which hand over data to authorities at their request, Telegram does not cooperate with authorities. This ensures a high degree of anonymity for its users which has proven beneficial for protest movements in authoritarian countries. This anonymity poses a barrier to political repression and criminal prosecution. This makes the service attractive not only for social

movements, but also for criminal and terrorist activities, and also facilitates the spread of hate-speech and calls for violence like those mentioned above.

The evolvement of the COVID-19 protests in Germany clearly demonstrate how online and offline contexts influence and often reinforce each other. But even though non-governmental organizations such as e.g. CEMAS had warned about the antidemocratic and violent potential of parts of the movement, and had pointed out the involvement of political activists from Germany's extremist far-Right, German authorities underestimated the dangerous potential of this movement for a long time. Only after journalists had revealed the plans to murder Michael Kretschmer, and after several gatherings in front of private homes of politicians had taken place, the issue became more of a serious concern.

This points out the relevance of investigative work by critical journalists and non-governmental organizations to monitor the mobilization and networking strategies of movements such as 'Querdenken' in Germany. However, while these actors often have to deal with large amount of data, their investigation often relies on manually conducted research and data evaluation. Machine Learning-based approaches for Natural Language Processing (NLP) are often not easily accessible for these actors, making it hard for them to process large amounts of data from the often very dynamic contexts of informal political networks.

1.2 Objective

This thesis aims at exploring the potentials of a particular NLP technology, namely Named Entity Recognition (NER) for the investigation of the mobilization for protests against COVID-19 measures in a German-language Telegram channel. The overarching goal is to explore if a component for Named Entity Recognition could be a beneficial contribution to the Open Source project 'Teledash', to make NER available as a tool for journalists, NGOs, and researchers investigating the COVID-19 related protest movements and networks in Germany.

- approach of NER in German-language Telegram messages
- extraction of entities particularly relevant for mobilization / monitoring of potential targeting: PERSON, ORGANIZATION; DATE; TIME; LOCATION; ACTION as custom type
- explore potentials and limitations of existing models
- explore potentials of translation approach
- explore potentials of transfer learning with Transformer models

1.3 Thesis Structure

- Theory
- Exploration
- Implementation
- Conclusion and Discussion: suggestions for future work

Chapter 2

Named Entity Recognition: An Introduction

Named Entity Recognition (NER) is the task of extracting named entities such as for example names of persons, locations, or organizations from text. The task is situated within the field of Natural Language Processing (NLP), and often an an important subtask of Information Extraction (IE) .

The origins of IE date back to the late 1970s, and its development was further accelerated in the late 1980s through funding by the US government for activities to evaluate IE technology. Organized by the Defense Advanced Research Projects Agency (DARPA) the first Message Understanding Conference (MUC) took place in 1987, providing standard datasets for international researchers to compete on. The MUC and was held seven times between 1987 and 1997. The first NER task at the MUC was introduced in 1996 and aimed at developing domain-independent systems for recognizing entities of types person, organization, and location, but also numerical expressions such as time, currency, and percentage (cf. Grishman and Sundheim 1996) in newspaper articles. In the following year, the organizers of the conference developed annotation guidelines which were used as a basis for several later NER tasks (cf. Chinchor 1997). Later activities such as the Automatic Content Extraction (ACE) meeting and the Text Analysis

Conference Knowledge Base Population (TAC KBP) shifted the focus to the extraction of more fine-grained entities, entity relations, and events, and later on to entity attribute extraction and entity linking. The provided datasets consisted of broader news data, covering political and international events, and later of open domain data, mainly from webpages (cf. Zong, Xia, and Zhang 2021, p. 229).

2.1 NER as a Sequence Labeling Task

NER is usually regarded as a sequence labeling task: To each word x_i in an input sequence, a label y_i is assigned, resulting in an output sequence Y which has the same length as the input sequence X (cf. Jurafsky and Martin 2021a, p. 148).

Since named entities often consist of more than one token, the task actually requires labeling spans of texts, which makes it necessary to determine the correct boundaries of an entity in a text, and assign the correct label to each token in a detected span (cf. *ibid.*, pp. 154–155).

NER thus involves the following two subtasks (cf. Zong, Xia, and Zhang 2021, p. 229):

1. **Entity detection**, to detect whether a given string in a text document is an entity, including the detection of the entity’s boundaries.
2. **Entity classification**, to assign a specific category to the detected entity.

2.2 Standard Named Entity Types

The types of entities extracted via an NER system differ depending on the domain and aim of the application or task. For biomedical NER systems, the entity types of interest may for example be names of biological proteins or diseases. There are however standard entity types which are commonly the focus of general NER approaches. These are **PERSON**, **LOCATION**, **ORGANIZATION**,

and sometimes **GPE** (geo-political entity). Often, an additional type is added which subsumes **OTHer** or **MISC**cellaneous named entities.

As we have seen above, earlier NER approaches also included numerical expressions such as time, date, currency/quantity, and percentage. Considering that a named entity is commonly defined as a single, unique real-world object, these numerical entities are not named entities in a narrow sense. They can, however, be useful categories depending on the purpose of an application (cf. Aggarwal 2018, p. 386).

2.3 The BIO Tagging Scheme

While some of the early NER approaches were exclusively rule-based, contemporary approaches usually rely on machine learning models trained on labeled datasets. For creating such datasets, an appropriate label set needs to be defined. A widely used tagging scheme is the BIO label set (cf. Ramshaw and Marcus 1995). Here, the B-tag is used to label the token representing the beginning of an entity (e.g. **B-PERSON**), the I-tag is assigned to all of the following (intermediate or final) tokens of the respective entity (e.g. **I-PERSON**). All tokens who are located outside of named entity spans are labeled with the **O** tag (cf. Zong, Xia, and Zhang 2021, p. 231). The corresponding tag set thus consists of two labels per entity and the **O** tag, resulting in a total of $2n + 1$ tags, with $n = \textit{number of entity types}$.

2.4 Evaluation of NER Systems

Similar to other NLP tasks, NER systems are usually systematically evaluated. This evaluation can be done in different stages of the development process, during deployment, and in production.

Evaluation in production usually involves application-specific business metrics, but also the involvement of stakeholders, and sometimes even end-users, which makes it rather expensive. Evaluation during development and deployment is therefore usually based on standardized machine learning metrics.

This intrinsic evaluation is less cost-effective since it can be done by the machine learning engineers themselves. Furthermore, it allows for adjustments and improvements in the earlier stages of the development process (cf. Vajjala et al. 2020, p. 71).

For supervised machine learning systems, the evaluation usually measures the model’s performance on unseen data, called the test set (cf. Jurafsky and Martin 2021a, p. 35). This test set is excluded from the training process and then used to evaluate the model. This is done by comparing the predictions of the model against the labels in the test set. (cf. Zong, Xia, and Zhang 2021, p. 241). The specific metrics used for intrinsic evaluation differ depending on the specific task. For NER, the standard metrics are F1, precision, and recall.

2.4.1 F1, Precision, and Recall

Precision and recall evaluate the model’s predictions by taking into account the number of entities correctly recognized by the model as named entities (true positives, TP), the number of entities falsely tagged as entities by the model (false positives, FP), and the number of named entities not detected by the model (false negatives, FN). Precision and recall are defined as follows ((cf. *ibid.*, p. 241):

$$precision = \frac{TP}{TP + FP} * 100$$

$$recall = \frac{TP}{TP + FN} * 100$$

Precision thus measures the percentage of correctly identified entities out of all entities detected by the system, i.e. how many correct predictions the model makes in its positive predictions. Recall measures the percentage of items tagged as named entities in the test set which are correctly recognized by the system, i.e. how many actual entities the model misses in its predictions.

The F1 score combines these two scores by computing a harmonic mean ((cf. Zong, Xia, and Zhang 2021, p. 241):

$$F1 = \frac{2 * precision * recall}{precision + recall}$$

F1 is the score commonly used in shared tasks on NER to evaluate the performance of the participating systems (cf. *ibid.*, p. 241).

Compared to other sequence labeling tasks, NER poses certain challenges for evaluation: The fact that named entities can consist of multiple tokens brings up a segmentation problem. For evaluation this means for example that a system which recognizes only one token of a person’s name consisting of two tokens will cause two errors: A false positive for tagging the second token as 0 instead of I-PERSON, and a false negative for not labeling the second token as I-PERSON) (cf. Jurafsky and Martin 2021a, p. 167).

Past shared tasks for NER have applied different strategies to evaluate the participating systems. Earlier tasks relied on relaxed metrics, considering a prediction as correct if the correct label was assigned, regardless of the correct boundaries, or if the correct text boundaries were detected, regardless of the correct label (cf. Yadav and Bethard 2019). Later, strict metrics were introduced in which a prediction was only considered correct if both the predicted label and boundaries of an entity matched the true label and boundaries (cf. *ibid.*).

2.4.2 Accuracy

Accuracy is another common machine learning metric. It measures the percentage of items correctly classified by the system:

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} * 100$$

However, accuracy is not suitable for classification tasks on highly imbalanced

datasets: In an NER dataset, the majority of tokens will be tagged as `O`. A classifier which would predict this majority class for all tokens without having learned anything from the training data would still achieve a very high accuracy (e.g. 90% accuracy for a dataset containing 90% `O` tags) (cf. Jurafsky and Martin 2021a, pp. 65–66). This is why accuracy is not a suitable metric for NER.

2.5 General Challenges for NER

As an NLP task, NER faces challenges resulting from the characteristics of human language.

Ambiguity, i.e. uncertainty or inexactness of meaning, is inherent to most human languages. Resolving the ambiguity of words or phrases usually requires knowledge of the context in which they appear. For example, the word ‘Bill’ might refer to a person’s first name, but also by a synonym for ‘invoice’ when located at the beginning of the sentence. Making computer-based systems take into account and ‘understand’ **context** is one of the major challenges for NER (cf. Aggarwal 2018, p. 386).

Humans furthermore constantly use **common knowledge** in their communication, meaning that facts are assumed to be known and thus not expressed explicitly in statements. While every language is constituted and shaped by rules, it is far from being exclusively rule-driven. Rather, languages are **creative** and heterogeneous, characterized by various styles, genres, and dialects. (cf. Vajjala et al. 2020, p. 14). Moreover, language evolves over time, in interaction with social and cultural processes, and the development of communication technologies such as social media. Existing datasets thus might not reflect new language phenomena such as hashtags. For named entities, this means that the set of known entities is never constant nor complete (Aggarwal 2018, p. 386).

Another challenge are **abbreviations** referring to multiple instances of the same named entity, such as ‘U.S.’, ‘USA’ or ‘United States of America’,

all referring to the same geopolitical entity. Entity disambiguation is thus usually an important follow-up task for NER(cf. Aggarwal 2018, p. 387).

Capitalization conventions vary widely across languages: In English-language texts for example, capitalization usually indicates proper names and can thus be used as a feature for NER. In German by contrast, all nouns are capitalized, which makes the feature less helpful (cf. Benikova, Biemann, and Reznicek 2014).

Diversity across languages makes it difficult to transfer a language-specific application to other languages. Building language-agnostic solutions is a very complex matter, while building separate solutions for each language is labor- and time-intensive (cf. Vajjala et al. 2020, p. 14).

Imbalanced resources for different languages limit the diversity both of research and applications. In fact, most NLP technologies are still designed for English or other European languages (cf. Singh 2008). Correspondingly, a large number of datasets and libraries exists for these high-resource languages. Low-resource languages are usually significantly ‘less studied, resource scarce, less computerized’ (Magueresse, Carles, and Heetderks 2020). This imbalance is also reflected in existing datasets for NER.

2.6 Challenges for NER in User-Generated Texts

While the above-mentioned challenges apply in general for all NER approaches, the characteristics user-generated texts from online domains further complicate the task for classic machine learning based NER approaches.

Social media texts are often **more informal and noisy** than newspaper articles, they contain misspellings or domain-specific expressions such as hashtags. Figure 2.1 illustrates the noisy character of tweets as an example.

Moreover, user-generated texts from social media are often of **shorter length** than e.g. newspaper articles which might result in a lack of context (cf. Rit-

ter et al. 2011).

The **wide variety of capitalization styles** are another challenge, like e.g. the use of only lowercase for German texts, or the use of uppercase only in some texts (cf. Ritter et al. 2011). As *ibid.* point out, earlier news-trained NER systems seemed to rely heavily on capitalization which limited their applicability to user-generated social media texts.

Also, these texts usually contain more out of vocabulary (OOV) words than e.g. news articles. This is partly due to misspellings, but also stems from platform specific codes and styles: The frequent occurrence of **special characters** such as # for hashtags or @ for user mentions poses challenges because they cannot easily be handled during annotation and training. User-generated text in social media also often includes a lot of **lexical variations**. *ibid.* collected more than fifty variations for the word ‘tomorrow’ from a Twitter corpus, as shown in figure 2.2.

1	The Hobbit has FINALLY started filming! I cannot wait!
2	Yess! Yess! Its official Nintendo announced today that they Will release the Nintendo 3DS in north America march 27 for \$250
3	Government confirms blast n nuclear plants n japan...don't knw wht s gona happen nw...

Figure 2.1: Examples of noisy text in tweets, taken from Ritter et al. 2011

‘2m’, ‘2ma’, ‘2mar’, ‘2mara’, ‘2maro’, ‘2marrow’, ‘2mor’, ‘2mora’, ‘2moro’, ‘2morrow’, ‘2morr’, ‘2morro’, ‘2morrow’, ‘2moz’, ‘2mr’, ‘2mro’, ‘2mrrw’, ‘2mrw’, ‘2mw’, ‘tmmrw’, ‘tmo’, ‘tmoro’, ‘tmorrow’, ‘tmoz’, ‘tmr’, ‘tmro’, ‘tmrow’, ‘tmrrow’, ‘tmrrw’, ‘tmrw’, ‘tmrww’, ‘tmw’, ‘tomaro’, ‘tomarow’, ‘tomarro’, ‘tomarrow’, ‘tomm’, ‘tommarow’, ‘tommarrow’, ‘tommoro’, ‘tom-morow’, ‘tommorrow’, ‘tommorw’, ‘tomm-row’, ‘tomo’, ‘tomolo’, ‘tomoro’, ‘tomorow’, ‘tomorro’, ‘tomorrrw’, ‘tomoz’, ‘tomrw’, ‘tomz’

Figure 2.2: Lexical variations of the word ‘tomorrow’ in tweets, taken from Ritter et al. 2011

Last but not least, social media texts often **mixed-language texts**, making it harder for language-specific models to accurately determine named entities in them.

Given these differences between social media texts and more formal genres, NER on social media needs to handle the problem of **domain adaptation**. Most of the existing NER systems are in fact trained on a small selection

of standard corpora mainly consisting of newswire texts, and earlier studies found that applying such a system to social media texts resulted in a drop of performance (cf. Ritter et al. 2011). This might also be true for the application of existing solutions to Telegram messages.

Chapter 3

Related Work

To the best of my knowledge, there are no works using NER for monitoring the COVID-19 protests in Germany, or any other works focussing on NER in German-language texts from messenger services. There are a few works using NER for analyzing (English-language) discourses in the context of the COVID-19 pandemic: Tangherlini et al. as well as Shahsavari et al. used NER as a component for analyzing conspiracy narratives and the corresponding communities in blog texts and news articles (Tangherlini et al. 2020), and in comments from the imageboard 4chan (Shahsavari et al. 2020).

There is, however, a number of shared tasks and corresponding datasets contributed to the development of German-language NER and NER in social media text data. For NER in social media texts, I will also present a selection of experimental studies which formed the basis for later shared tasks.

3.1 Shared Tasks on German-Language NER

A first important cornerstone for German-language NER was the Conference on Computational Natural Language Learning (CoNLL) for English and German-language NER in 2003. The task focused on the entity types person, location, organization, and miscellaneous. The provided German dataset consists of 909 newspaper articles from the Frankfurter Rundschau.

The dataset was annotated by non-native speakers at the University of Antwerp which is assumed to have led to annotation inconsistencies.¹ The best participating system was a system combining Maximum Entropy and Hidden Markov Models, and achieved an F1 score of 72.41 for the German-language dataset (cf. Tjong Kim Sang and De Meulder 2003)

In 2014, GermEval, a series of shared tasks on Natural Language Processing for German, announced an NER task. The provided dataset consists of 31,000 sentences sampled from Wikipedia articles and online news, and includes fine-grained annotations for entities of type person, organization, location, and other. The Germeval 2014 corpus includes nested and derived entities and is thus not readily combinable with other datasets. While the corpus consists of online resources, it does not contain noisy user-generated texts. The best system achieved an F1 score of 79.08 for first-level named entities, using character query-based features in order to find sequences that are characteristic for named entities, and vocabulary clusters for semantic generalization (cf. Benikova, Biemann, and Reznicek 2014).

The two corpora represent standard datasets for German-language NER and are used until today to train and evaluate state of the art NER models.

3.2 Experimental Studies and Shared Tasks on Social Media NER

While the first shared tasks on NER focussed on newswire articles, later work began to explore NER in user-generated texts from social media. Research on NER for social media texts is almost completely focused on English-language texts: In 2017, Bhoi and Balabantaray analyzed seven different systems for NER in social media text, out of which five were exclusively English. The only system supporting German language was the Stanford NER tagger, which was trained on newswire, not on social media texts though. The

¹Some authors, e.g. Schweter and Akbik 2020 mention a revised edition, but I was not able to find further details or the mentioned resource itself.

3.2. EXPERIMENTAL STUDIES AND SHARED TASKS ON SOCIAL MEDIA NER¹⁷

authors compare system performance for tweets and microblog content. They report best $F1=0.686$ for tweets, and best $F1=0.889$ for the more formal microblog content, concluding that ‘state-of-the-art NER methods are giving unsatisfactory results in that informal and noisy content of social media text’ (Bhoi and Balabantaray 2017)

Ritter et al. were among the first to explore NER in social media texts. In 2011, the authors conducted an experimental study, using a random sample of 2,400 English-language tweets annotated with ten different entity types². The authors use a discriminate approach for segmentation, and a weakly supervised approach for classification, relying on topic modeling and a large knowledge base. The system achieved an $F1$ score of 0.66 when applied to ten entity types. When constrained to the three standard entity types person, location, and organization, the $F1$ score increased to 0.75, with 0.83 for person, 0.74 for location, and 0.66 for organization (cf. Ritter et al. 2011).

Another study conducted in 2015 by Derczynski et al. 2015, using three different English-language datasets including the Ritter et al. corpus, confirmed the rather low performance of by then existing machine-learning based NER on noisy user-generated data: The evaluated systems achieved $F1$ scores ranging between 40.27, 51.49, and 77.18 on the different datasets (cf. *ibid.*). In line with Ritter et al. 2011, the authors identify unreliable capitalization, typographic errors, abbreviations, and lack of context due to the shortness of texts as main challenges.

Also in 2015, Baldwin et al. organized a shared task for named entity recognition on social media data, namely tweets from Twitter in the context of the Workshop on Noisy User-generated Text (W-NUT), using the Ritter et al. corpus. The best participating system combined the Stanford NER tagger with entity linking via a knowledge base. It achieved an $F1$ score of 56.41 (cf. Baldwin et al. 2015). The W-NUT series continued to address the task of NER in subsequent years: Tasks were organized on named entity recognition in tweets (cf. Strauss et al. 2016), and novel and emerging entity recognition

²Including person, geo-location, company, product, facility, tv-show, movie, sportsteam, band, and other (cf. Ritter et al. 2011)

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(cf. Derczynski et al. 2017), both using the Ritter corpus.³ The achieved F1 scores remained rather low with best F1=52.41 in 2016, and best F1=41.86 in 2017.

Even though NER on social media texts remained a challenging task regarding these results, W-NUT shifted its focus to other tasks and domains in subsequent years. In parallel, larger human-annotated social media NER datasets have been created (cf. Derczynski, Bontcheva, and Roberts 2016). However, there is still a lack of such corpora for languages other than English.

While the introduction of transformer models in recent years has outperformed prior approaches in many NLP tasks including NER, they are commonly fine-tuned and evaluated on the standard CoNLL and Germeval 2014 corpora.

³In 2017 the Ritter corpus was used as training set, extended by ~ 1.000 manually annotated documents from different online forums and platforms as development set, and ~ 1.300 documents manually annotated Youtube comments as test set

Chapter 4

Transformer Models: State of the Art for NER

Since their beginnings in the 1970s, NER approaches have evolved mostly in parallel to general advancements in NLP: Starting with early rule-based and knowledge-based systems, the field moved to machine-learning based approaches, often involving extensive feature-engineering. Deep-learning based approaches using neural networks and word embeddings then began to replace these architectures, making feature-engineering more and more obsolete. Most recently, transformer models based on self-attention have achieved new state of the art performance for several NLP tasks including NER.

The transformers architecture was first introduced in 2017 by Vaswani et al., in their seminal paper ‘Attention Is All You Need’, in which the authors propose a ‘new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.’ (Vaswani et al. 2017)

Transformers thus represent a clear shift from prior model architectures. Like the architectures described above, transformer models map input sequences to output sequences of the same length. This sequence-to-sequence architecture is also called encoder-decoder architecture (cf. Tunstall et al. 2022,

p. 3).

But unlike prior approaches, transformers do not follow a sequential pattern for modelling context. Rather, they use a mechanism called **self-attention** which allows them to represent each word with respect to its current context (cf. Vajjala et al. 2020, pp. 25–26).

Through self-attention, these models are able to use information from (theoretically) arbitrarily large contexts, without having to pass it sequentially through recurrent connections as in an RNN (cf. Jurafsky and Martin 2021b, p. 17).

When processing an input item, the self-attention layer has access to the item itself and all of the inputs before it, but not to the succeeding inputs.¹ The computation for each item is carried out independently, which allows to run a high number of computations in parallel (cf. *ibid.*, p. 18). Transformer models thus allow for parallelization which is an advantage in terms of computational speed and efficiency compared to prior approaches.

Attention in transformer models can be conceptualized as ‘[comparing] an item to a collection of other items in a way that reveals their relevance in the current context.’ (*ibid.*, p. 18). Self-attention then means to compare a given item to other elements within a sequence (cf. *ibid.*, p. 18).

In transformer models, each input embedding is considered in three different roles (cf. *ibid.*, p. 19:

- As a **query**, the input embedding represents the current focus of attention and is compared to all of its preceding inputs.
- If the input embedding is among the preceding inputs, it is compared to the current query (focus of attention). In this case, it represents a **key**.
- Whenever we use the embedding for computing the current output, it

¹This only applies to unidirectional, backward looking self-attention layers. In bidirectional architectures such as BERT, the considered context is not limited to the preceding inputs. See Devlin et al. 2018 for further explication.

is referred to as **value**.

Figure 4.1 illustrates the process of calculating the value for the third element in a sequence including the key, query, and value vectors for the current item x_3 and the preceding items x_1 and x_2 . We can see that after generating these vectors, they key-query comparisons are carried out, which are then scaled using the softmax function. The resulting values are then multiplied by the value vector, and summed up to obtain the output vector, representing the output vector y_3 .

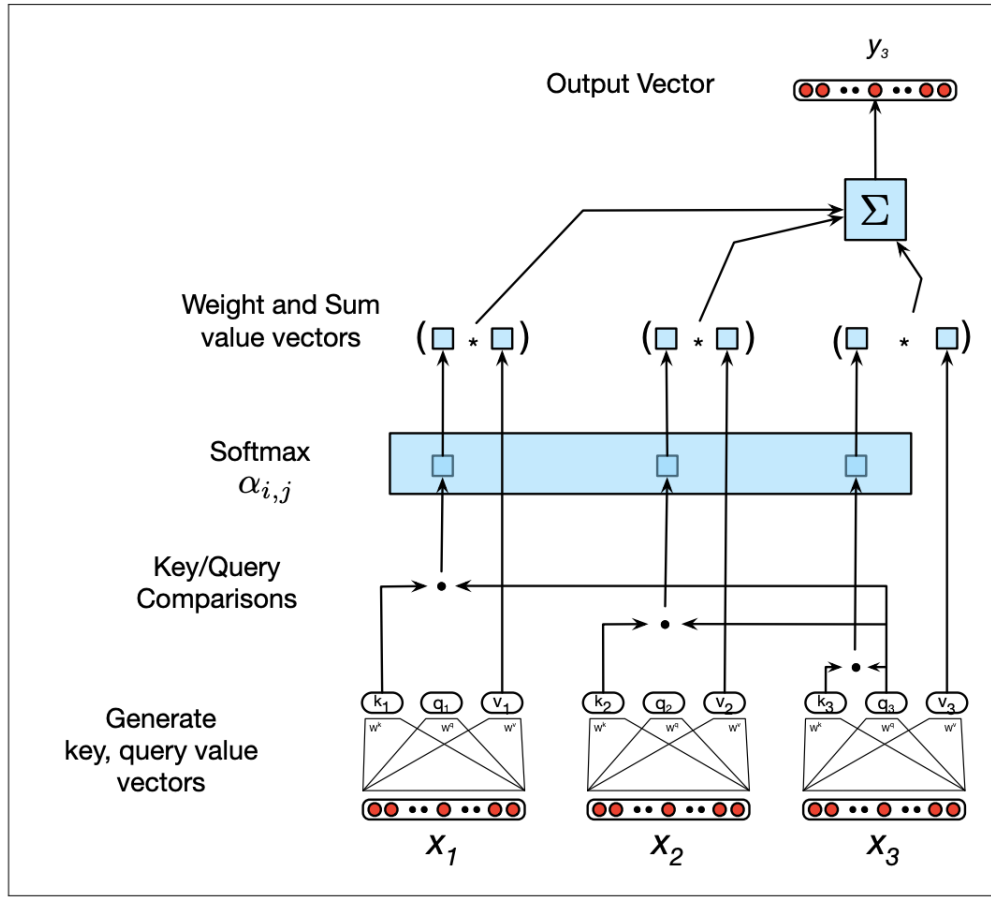


Figure 4.1: Calculating the value y_3 for the third element x_3 in a sequence (using left-to-right-attention); taken from Jurafsky and Martin 2021b, p. 20

The way context is conceptualized and modelled in this architecture is thus fundamentally different than in sequential models. By extending the concept

of self-attention to multiple heads, transformer models are furthermore able to consider different aspects of relationships between words at the same time. This mechanism is called multi-head attention, and implemented in the form of multiple, parallel self-attention layers which are located at the same depth of the network. Each of these heads learns its own parameters, allowing them to pay attention to different ways in which the words in the input relate to each other (cf. Jurafsky and Martin 2021b, p. 23).

Transformer models are composed of so-called transformer blocks. All of them have the same input and output dimensions, which allows to stack them on top of each other (cf. *ibid.*, p. 22).

These transformer blocks are themselves multilayer networks, containing the self-attention layers, simple linear layers, and feedforward networks. As a mechanism to improve training performance, residual connections and layer normalization are implemented in each block, allowing to pass information directly to the next layer, and keeping the output values of each hidden layer in a certain range (cf. *ibid.*, p. 22).

The great advantages of transformer models compared to sequential models are thus that they allow for parallelization and that the model context in a more complex, multi-faceted way. However, the way attention is computed makes the process expensive for long inputs, since for every input, each pair of tokens needs to be compared. This is why most models are in practice limited to a certain input length (cf. *ibid.*, p. 21).

Furthermore, since transformer models do not work sequentially, the information about the positional context of each item in the input is lost. To provide this information, transformer models use additional positional embeddings to include this information. These embeddings are learned during training, just like the word embeddings, and encode information about the context for each input item (cf. *ibid.*, pp. 23–25).

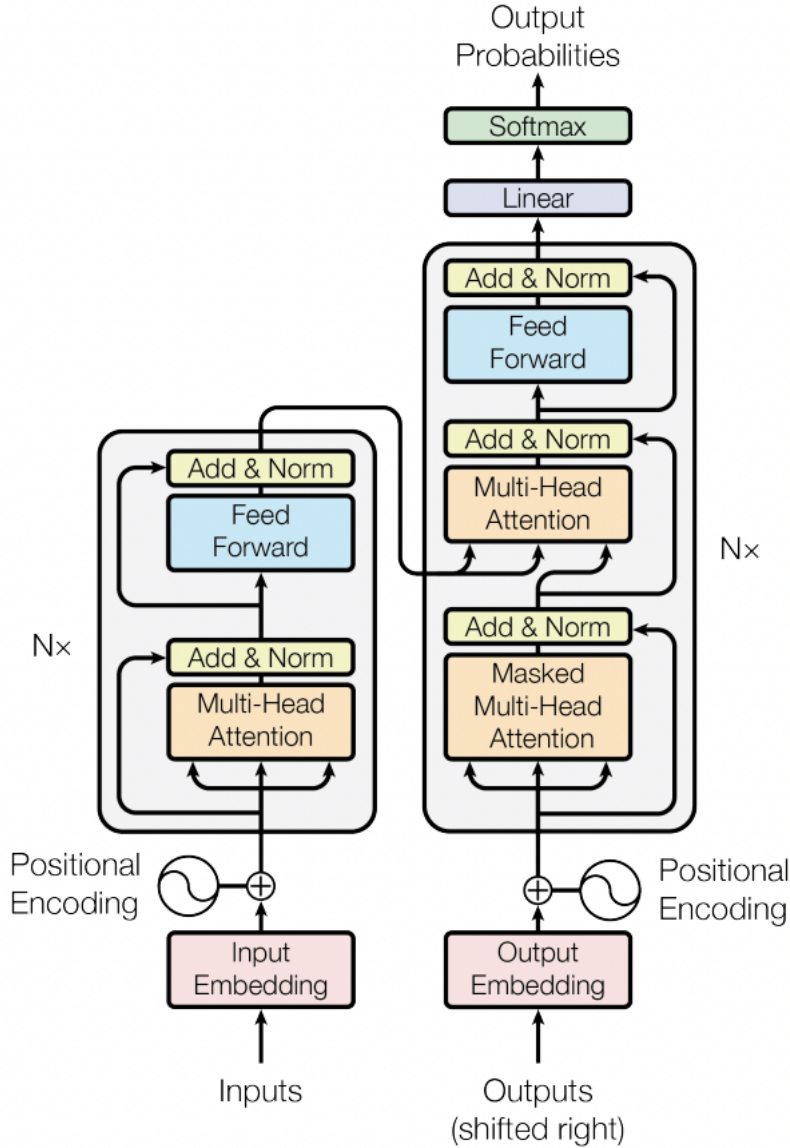


Figure 4.2: Transformer model architecture as introduced by Vaswani et al. 2017

Transformers are usually trained to serve as large language models, on different tasks such as next sentence prediction (NSP) or masked language modeling (MLM). They can also be used for sequence labeling tasks such as NER. This is achieved by first training a language model on large amounts of unlabelled text data, which is called pre-training. Pre-training allows the model

to incorporate rich information about language in general, which allows it to easier learn specific tasks, called downstream tasks, on smaller datasets later. This second step is called fine-tuning ((cf. Jurafsky and Martin 2021b, pp. 27–28). Fine-tuning requires labeled data, but can be achieved with datasets of much smaller size, since the model already learned a lot about a given language during pre-training.

There is a large number of different architectures for transformer models, resulting in an even larger number of available models.²

The three main architectures for transformer models are encoders, decoders, and encoder-decoders. Encoder models are still the most commonly used, both in research and for business applications (cf. Tunstall et al. 2022, p. 79), and the models selected for the experiments here as well rely on the encoder architecture. These general architecture types can be further differentiated into different model architectures such as e.g. DistilBERT or XLM models.³

Models which have been pretrained on different corpora and can be used e.g. for fine-tuning tasks are called checkpoints. The term 'model' is thus a general term which can refer to either a a specific architectural model skeleton or a checkpoint of a pretrained model (cf. HuggingFace 2022c).

²By the time of writing this thesis, the Hugging Face Hub lists more than 62.000 of pretrained models available for download: <https://huggingface.co/models>, accessed on 2022/08/06.

³See Tunstall et al. 2022, p. 79 for an overview and *ibid.*, pp. 57–86 for details on the different architectures.

Chapter 5

SpaCy and Flair: A Qualitative Exploration of Existing Solutions

Most of the existing NER models are limited to the four common entity types PERSON, LOCATION, ORGANIZATION, and OTHER/MISC. To the best of my knowledge, there are no existing models supporting the detection of the entity types TIME, DATE, and ACTION. An exception is the English-language spaCy model which supports TIME and DATE recognition.

In order to explore the applicability of existing models for my purpose, I selected some sample messages from a German-language Telegram channel and qualitatively evaluated the results of two German-language NER models. Since the English spaCy model supports more entity types, and English models are known to be well-resourced, I used machine translation to translate the sample texts into English and explore the results of the English models as well.¹

All tested models were selected from the library spaCy and the framework

¹For implementation details, see https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/3_exploration/spacy_flair_bert_first_experiments.

FLAIR. The models were chosen due to their easy applicability, allowing for fast exploration and analysis, and because both of them support German- and English-language named entity recognition. I am aware of the fact that the observations presented here cannot be generalized without quantitative evaluation on a larger sample.

5.1 Example Data

I randomly selected ten example messages from the mainly German-language Telegram channel ‘Demotermine’ (*Demo dates*). Prior qualitative exploration of sample messages indicated that this channel contains a lot of messages mobilizing for various kinds of protests against the COVID-19 containment measures in Germany. This makes it a suitable source for my purpose.

5.2 German-language NER with spaCy and Flair

My first analysis focusses on results for German-language sample texts, using the medium-size German-language spaCy model and the standard German-language Flair NER model.

5.2.1 The German-language spaCy model

spaCy is a free open-source library for NLP. The library is extensively documented and thus serves as an easy starting point for beginners, while also claiming to be designed for use in production. spaCy supports more than 66 languages (including German) and supports a variety of NLP tasks such as part-of-speech tagging, text classification, and NER (cf. ExplosionAI 2022b). The medium-size German model supports to the common entity types LOC, MISC, ORG, and PER (cf. ExplosionAI 2022c).

The central component of the model’s NER pipeline is spaCy’s Entity Recognizer based on transition-based algorithm which assumes that the most

relevant information about an entity is close to its first token (cf. ExplosionAI 2022a). The underlying `TransitionBasedParser` is a neural network state prediction model which embodies ‘an approach to structured prediction where the task of predicting the structure is mapped to a series of state transitions.’ (ExplosionAI 2022d).

The model was trained on the WIKINER corpus, a free, multilingual corpus created from Wikipedia articles. The annotations were created automatically (cf. Nothman et al. 2013). The corpus is thus only silver standard (compared to gold standard corpora based on human annotations). The model achieved an F1 score of 0.84 (cf. ExplosionAI 2022c).

5.2.2 The German-language Flair Model

The FLAIR framework is an NLP library developed to ‘facilitate training and distribution of state-of-the-art sequence labeling, text classification and language models’ (Akbik et al. 2019). The overall aim is to provide an interface for the easy use of different word and document embeddings (cf. *ibid.*).

The German-language Flair model for NER uses a combination of GloVe embeddings (cf. Pennington, Socher, and Manning 2014 and Flair embeddings (cf. Akbik, Blythe, and Vollgraf 2018) to create word embeddings. Flair embeddings create ‘contextual string embeddings’ (*ibid.*) which generate different embeddings for the same word depending on its context in a document (cf. *ibid.*). The word embeddings passed to a Bidirectional Long Short Term Memory (BiLSTM)-Conditional Random Field (CRF) sequence labeling model for the NER downstream task (cf. *ibid.*).

I used Flair’s German standard 4-class NER model, supporting the entity types person, location, organization, and other. It was trained on the CoNLL 2003 corpus, reaching an F1 score of 87.94 (cf. Akbik, Blythe, and Vollgraf 2022b).

5.2.3 Results

Table 7.1 shows the results for both models out of which the following observations can be made: Overall, the spaCy model yields more false positives: It falsely tags the sequences ‘Endkundgebung’ (final rally), ‘GEHT DEMONSTRIEREN’ (go demonstrate), ‘Dresdner &’, ‘Videos’, ‘Berliner’, ‘Versammlungsfre’ (freedom of assembly, truncated), ‘Einfach’ (simple/simply), ‘Der Mann mit dem Schild’ (the man with the sign), and ‘Einsatzwagen’ (patrol cars).

The Flair model only falsely tags ‘FU’ and ‘Demo-Berlin’. The sequence ‘FU’ does not exist in the original text, but is apparently a result of the model’s pre-processing. Tagging ‘FU’ as organization is plausible however, as is tagging ‘Demo-Berlin’ as location since the sequence contains ‘Berlin’.

The spaCy model sometimes correctly detects a named entity, but then assigns the wrong tag to it: ‘Blankenfelde Mahlow’ represents a location, but is tagged as person.

The spaCy model does not detect ‘Freiburg’ in the first message and thus yields a false negative, while the Flair model correctly detects the location mention.

Somewhat surprisingly, spaCy recognizes parts of the sequence ‘Platz der alten Synagoge’ (square of the old synagogue) as a location which is missed by the Flair model.

Even though the sample is too small for generalization, my observations casts strong doubts on the claim that spaCy offers a production-ready NER solution. I assume that the differences in performance result both from the different model architectures and the quality of the respective training corpora.

Table 5.1: Detected entities for German-language example messages

No.	Text	spaCy/de_core_news_md	flair/ner-german
1	Freiburg Aufzug mit Endkundgebung am Platz der alten Synagoge . Live Musik ! Details 19062021 Raus auf die Straße	Endkundgebung[LOC], alten Synagoge[LOC]	Freiburg[LOC]
2	GEHT DEMONSTRIEREN ! - Message von Elijah Tee an die Dresdner & anderen Demonstranten in Deutschland Morgen am 13.06. um 16.30 Uhr und bundesweit überall , wo es Mitdenker gibt . - Bilder und Videos hochladen - Diskussion Fahren oder Mitfahren	GEHT DEMONSTRIEREN [ORG], Elijah Tee[PER], Dresdner &[LOC], Deutschland[LOC], Videos[MISC]	Elijah Tee[PER], Deutschland[LOC]
3	15831 Blankenfelde Mahlow Berliner Speckgürtel Ausfahrt Lichtenrade am 14.03.2021 14032021 Raus auf die Straße	Blankenfelde Mahlow[PER], Berliner[MISC]	Blankenfelde Mahlow[LOC], Lichtenrade[LOC]
4	2021 10 11 Sven Liebich fragt Ministerpräsident Haseloff nach dessen Verständnis von Versammlungsfrei - Sven Liebich 0:23 Raus auf die Straßen Übersicht / Overview	Sven Liebich[PER], Haseloff[PER], Versammlungsfrei[PER], Sven Liebich[PER]	Sven Liebich[PER], Haseloff[PER], Sven Liebich[PER]
5	FÜR ALLE DEMOKRATEN GEFAHR : Die Polizei blockiert den Stern ! Deswegen laufen die Menschen vom Ernst-Reuther-Platz nach Südosten zum Breitscheidplatz und dort weiter auf den Kudamm ! ! Schließt euch an : Neue Kantstraße direkt Richtung Osten auf den Breitscheidplatz ! Von der Straße des 17. Juni Richtung Süden zum Kudamm !	DEMOKRATEN[ORG], Ernst-Reuther-Platz[LOC], Breitscheidplatz[LOC], Kudamm ! [LOC], Neue Kantstraße[LOC], Breitscheidplatz[LOC], Kudamm ! [LOC]	FU[ORG], Ernst-Reuther-Platz[LOC], Breitscheidplatz[LOC], Kudamm[LOC], Kantstraße[LOC], Breitscheidplatz[LOC], Kudamm[LOC]
6	Dr. Gunter Frank : Wie Moralismus dazu führt das man Feinde sieht und vernichten will — Antihygienista - Offene Gesellschaft Kurpfalz 15:04 Raus auf die Straßen Übersicht / Overview	Gunter Frank[PER], Antihygienista[LOC], Offene Gesellschaft Kurpfalz[ORG]	Gunter Frank[PER], Antihygienista[ORG], Offene Gesellschaft Kurpfalz[ORG]
7	Einfach nur Krank solche Politiker ! Was sollen wir mit ihm machen wenn unsere Kinder anhand dieser Impfung nen schaden erleiden ? Wer ist dann dafür verantwortlich , wen schmeißen wir dann in den Knast und nehmen sein ganzes Geld weg , sollen wir das mit dir machen ? Raus auf die Straße	Einfach [MISC]	<i>no entities detected</i>
8	27.11.2021 Frankfurt Durchsagen von gemischt mit Musik . Super Idee Für mehr Berichte folgen auf ... Raus auf die Straßen Übersicht / Overview	Frankfurt[LOC]	Frankfurt[LOC]
9	Demo-Berlin : Der Mann mit dem Schild wurde soeben von der Polizei festgenommen . Circa 7 Einsatzwagen sind vor Ort und umzäunen den Park großräumig . Am Humboldthain gibt es nichts mehr zu gewinnen . # b2908 Raus auf die Straßen	Der Mann mit dem Schild[LOC], Einsatzwagen[LOC], Humboldthain[LOC]	Demo-Berlin[LOC], Humboldthain[LOC]
10	Anthony Fauci : Definition von “ vollständig geimpft ” könnte geändert werden — Epoch Times Nachrichten 6:59 Raus auf die Straßen Übersicht / Overview	Anthony Fauci[PER], Times Nachrichten[MISC]	Anthony Fauci[PER], Epoch Times Nachrichten[ORG]

5.3 Exploring the Potentials of Machine Translation for NER

Even though the German spaCy model performed worse than the German Flair model, its English equivalent comes with one great advantage for my purpose: It supports a greater number of entity types, including DATE and TIME, making it potentially useful for monitoring mobilization for protests. Furthermore, regarding the fact that English models are generally known to be more well-resourced, my aim was to explore the performance of the English variant for both models.

5.3.1 The English-language spaCy Model

The English-language spacy model supports a total of 18 entity types, including the standard entity types person, location, organization. For my aim, the entity types time and date are particularly relevant. For a detailed overview of all entity types and their definitions see table A.1 in the appendix.

The underlying architecture is the same as for the German model.

Judging by the listed data sources in the documentation, I assume that NER model was trained on the OntoNotes 5 corpus, containing data from various text genres including news, phone conversations, blogs, usenet newsgroups, etc. (cf. Ralph Weischedel 2013). The model achieved an F1 score of 0.85.

5.3.2 The English-language Flair Model

The English-language Flair model for NER supports the same four standard entity types as the German version. Unlike the English spaCy model, it offers no advantages in terms of entity types.

The architecture of the standard English-language Flair model is the same as for the standard German model described in section 5.2.2.

The English model was trained on a corrected version of the CoNLL 2003 corpus and achieved an F1 score of 93.06 (cf. Akbik, Blythe, and Vollgraf

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2022a).

For implementation, the example messages were first translated using the DeepL API.² The implementation for obtaining the NER results from both models followed the same procedure as for the German models.³

5.3.3 Results

Table 5.2 shows the results of both models for the translated texts.

We can observe that the overall performance of the spaCy model deteriorates compared to the German version: Out of the 25 entity tags, 17 are misclassifications, e.g. the person name ‘Sven Liebich’ as organization, or the location ‘Humboldthain’ as person, to name just a few. Furthermore, it misses named entities which were recognized by its German equivalent, e.g. the person mentions ‘Haseloff’ and ‘Elijah Tee’. The model also proves largely unsuitable for extracting dates and times: Misclassifying e.g. the date ‘13.06’ or the time mention ‘16.30’ as cardinals, and the postal code ‘15831’ as date. Out of the 11 mentions of times or dates in the example data, it correctly recognizes only the sequence ‘03/14/2021 14032021’ in message no. 3 as date.

The performance of the English Flair model worsens as well compared to its German equivalent, although not quite as strong as the spaCy model: The model yields more false entity classifications, e.g. of ‘Raus auf die Straßen Overview’ as organization, or ‘Dresdeners’, ‘Streets Overview’, ‘Stern!’ as miscellaneous. Regarding locations, the model now misclassifies the mentions in message no. 3 as miscellaneous, which were at least partly correctly tagged by the German Flair model. Interestingly, it is the only model which partly recognizes the location mention ‘Straße des 17. Juni’.

²<https://www.deepl.com/docs-api>. For implementation details of the translation step, see: https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/3_exploration/translation

³For NER implementation, see https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/3_exploration/spacy_flair_bert_first_experiments.

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Table 5.2: Detected entities for English-language example messages

No.	Text	spaCy/en_core_web_md	flair/ner-english
1	Freiburg procession with final rally at the square of the old synagogue . Live music ! Details 19062021 Out on the street	<i>no entities detected</i>	Freiburg[LOC]
2	GO DEMONSTRATE ! - Message from Elijah Tee to the Dresdeners & other demonstrators in Germany Tomorrow on 13.06. at 16.30 clock and nationwide everywhere , where there are fellow thinkers . - Upload pictures and videos - discussion drive or ride along	Germany[GPE], Tomorrow[DATE], 13.06[CARDINAL], 16.30[CARDINAL]	Elijah Tee[PER], Dresdeners[MISC], Germany[LOC]
3	15831 Blankenfelde Mahlow Berlin Speckgürtel Lichtenrade exit on 03/14/2021 14032021 Out on the road	15831[DATE], Blankenfelde Mahlow Berlin[PERSON], Speckgürtel Lichtenrade[PERSON], 03/14/2021 14032021[DATE]	"Blankenfelde Mahlow Berlin Speckgürtel Lichtenrade"[MISC]
4	2021 10 11 Sven Liebich asks Prime Minister Haseloff about his understanding of assembly free - Sven Liebich 0:23 Raus auf die Straßen Overview / Overview	Sven Liebich[ORG], Haseloff[PER], Straßen Overview / Overview[ORG]	Sven Liebich[PER], Haseloff[PER], Sven Liebich[PER], Raus auf die Straßen Overview[ORG]
5	FOR ALL DEMOCRATS DANGER : The police blocks the Stern ! That's why people run from Ernst-Reuther-Platz to the southeast to Breitscheitplatz and there further to Kudamm ! ! Join : Neue Kantstraße directly to the east to Breitscheitplatz ! From the Straße des 17. Juni southwards to the Kudamm !	DEMOCRATS[NORP], Ernst-Reuther-Platz[ORG], Breitscheitplatz[PERSON], Kudamm[FAC], Neue Kantstraße[PERSON], Straße[GPE], 17[CARDINAL], Kudamm[FAC]	Stern ![MISC], Ernst-Reuther-Platz[LOC], Breitscheitplatz[LOC], Kudamm[LOC], Neue Kantstraße[ORG], Breitscheitplatz[LOC], Straße des[LOC], Juni[LOC], Kudamm[LOC]
6	Dr. Gunter Frank : How moralism leads to the fact that one sees enemies and wants to destroy - Antihygienista - Offene Gesellschaft Kurpfalz 15:04 Raus auf die Straßen Übersicht / Overview	Gunter Frank[PER], 15:04 Raus auf[PER], Straßen Übersicht / Overview[ORG]	Gunter Frank[PER]
7	"Just sick such politicians ! What should we do with him if our children suffer from this vaccination ? Who is responsible for it , who do we throw in jail and take away all his money , shall we do that with you ? Out on the street"	<i>no entities detected</i>	<i>no entities detected</i>
8	27.11.2021 Frankfurt announcements of mixed with music . Super idea For more reports follow on ... Out on the streets Overview	Frankfurt[GPE]	Frankfurt[LOC]
9	Demo-Berlin : The man with the sign has just been arrested by the police . About 7 emergency vehicles are on site and fence the park extensively . At the Humboldthain there is nothing more to win . # b2908 Out on the streets	Humboldthain[PER], #[CARDINAL]	Humboldthain[LOC]
10	Anthony Fauci : Definition of " fully vaccinated " could be changed - Epoch Times News 6:59 Out on the Streets Overview / Overview.	Anthony Fauci[PER], the Streets Overview / Overview[ORG]	Anthony Fauci[PER], Epoch Times News[ORG], Streets Overview[MISC]

5.4 Summarization

My findings indicate that existing solutions are only of limited use for my purpose: The predictions of the spaCy model contain a lot of misclassifications, both for German and English. The Flair model delivers more accurate results, with a decrease in performance for the English data. One reason for this could be that the person or location names are language area specific, and therefore better recognized by models trained on data from their origin language area. Most importantly, most of the analyzed models are restricted to the standard entity types person, location, organization, and other/miscellaneous. This makes it impossible to extract e.g. time or date mentions which are of central relevance for monitoring COVID-19 protests. The only model which provides these entity types delivers poor results and misses by far most of the date and time mentions included in the examples. Against this background, the remaining part of my thesis explores the potentials and limitations of building a custom solution.

Chapter 6

Building a Custom Model for NER in Telegram Channels

As we have seen in the previous chapter, existing solutions are not applicable for my purpose. Hence I decided to train a custom NER model for extracting the entity types person, location, organization, time, date, and action from Telegram messages. For this, I fine-tuned a selection of pretrained transformer models using the Hugging Face Transformers library.¹ Since fine-tuning for NER requires labeled data, I created a detailed annotation scheme and annotated a small-size Telegram corpus of 500 messages. As additional data, I used the Smartdata corpus, a dataset consisting of 2.598 documents created for event extraction in online texts. I evaluated five different models on the two datasets separately, and on the combined corpus. Finally, I selected the best model regarding a combined assessment of its F1, precision and recall scores.

6.1 Building the Telegram Corpus

The small-size Telegram corpus was created from the same channel I used for my exploration of existing models in chapter 5: The channel ‘Demotermine’,

¹<https://huggingface.co/docs/transformers/>

disseminating mainly messages mobilizing for protests against COVID-19 measures in German language.

6.1.1 Raw Data

The raw data covers a time period from March 3rd, 2018 when the channel was created, until December 19th, 2021. By the beginning of June 2022, the channel had around 52.900 subscribers.

The channel thus existed long before the COVID-19 pandemic. However, an analysis of message frequencies over time clearly shows the channel gained significant momentum in spring 2020, at the beginning of the pandemic in Germany (see figure 6.1).

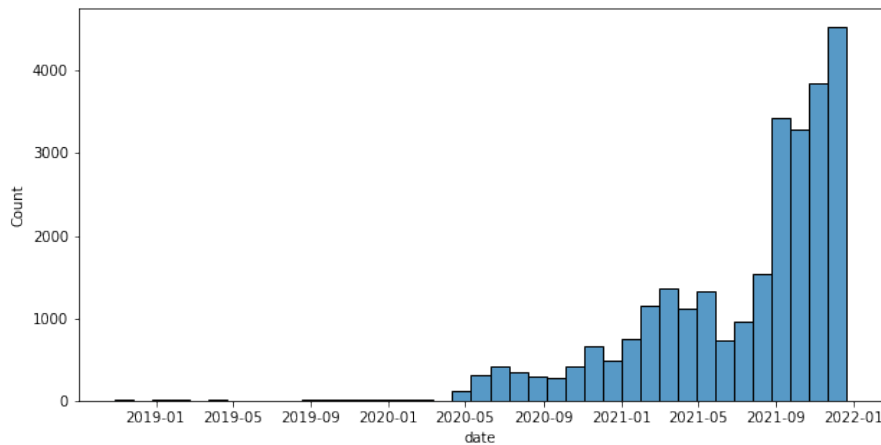


Figure 6.1: Message frequencies in Telegram channel ‘Demotermine’

6.1.2 Text Cleaning

Text cleaning was restricted to an absolute minimum to make the corpus reflect a realistic sample of user-generated texts. Besides the removal of newline characters, compound hashtags were split up into separate hashtags (e.g. transforming #food#delicious into #food #delicious).

@*user_mentions* and #*hashtags* were included as whole sequences, same as sequences containing in-word hyphens. URLs and emojis were included as well. The capitalization of words was retained. For tokenization, the tokenizer of the German medium-size spaCy model was applied with a customized token match pattern.² Text cleaning resulted in empty rows for some documents. After removing these empty rows and text duplicates, 27,627 messages remained for sample selection.

6.1.3 Sample Selection

The following steps were carried out for sample selection:³ To be able to focus on German-language messages, I used the `langdetect` package `langdetect` to all messages. Results show that German content is clearly the predominant language with $\sim 83\%$ (22,803 messages), followed by English ($\sim 3.8\%$, 3,823 messages), and then French, Italian, and Dutch with less than 1% each. Language detection was sometimes flawed for mixed-language texts. These were removed manually when encountered in the annotation process. After constraining the dataset to German-language texts, 22,803 messages remained, out of which texts with minimum length of 100 characters were selected, resulting in 14,605 remaining messages. Out of these, I selected twenty percent as a random sample, resulting in 4,021 messages. Since earlier findings in the context of our research project ‘Digitaler Hass’ revealed that Telegram channels often contain a rather high number of very similar messages, I removed such similar messages based on Levenshtein distance with a threshold of 0.8, resulting in 3,095 remaining messages.

While defining a minimum length for the sample seems reasonable, because the selected texts are likely to be comprehensible and contain a relevant number of entities, this constraint does not reflect the fact that social media texts are often of very short length. In fact, almost 50% of the messages were

²For implementation details, see https://github.com/eli-quereli/ner-german-telegram/blob/main/src/notebooks/preprocessing/v02_preprocessing.ipynb.

³For implementation details, see https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/data_selection.

shorter than 100 characters. To create a realistic sample, I included some of these short texts as well, using the same procedure as described above, except for the sampling size which was 50% instead of 20%. After random sampling and removing similar messages, 540 short messages remained and were added to the first sample.

6.1.4 Annotation

Data annotation turned out to be one the most challenging tasks of my project. First of all, existing annotation schemes for NER vary greatly in terms of the detailedness and precision of provided definition, the considered entity types, their granularity, and last but not least in how they handle social media-specific phenomena like user mentions, urls, or hashtags.⁴

I decided to create my own annotation scheme including entity type definitions, detailed instructions and a number examples to ensure consistent annotations also for ambiguous cases. The full annotation schema and guide can be found in the appendix (A). The tag set covers the six entity types person, location, organization, time, date and action and follows the BIO tagging scheme (see section ??). I used Labelstudio⁵ for annotation. The O-tag was assigned automatically to all tokens left unlabeled.⁶

6.1.5 Telegram Corpus Statistics

Due to limited time and resources, I did not annotate the whole sample. The final annotated corpus consists of 500 documents including 2,509 annotated named entities. Table 6.1 shows the distribution of the different entity types after the train-test-validation split. Locations are by far the most often mentioned entity types ($\sim 51\%$), followed by dates ($\sim 17\%$), and action mentions

⁴An extremely tentative example is the annotation guide for the W-NUT 2015 shared task on NER in tweets: https://docs.google.com/document/d/12hI-2A3vATMWRdsKkzDPHu5oT74_tGO-PPQ7VNOIRaw/edit.

⁵<https://labelstud.io/>

⁶For implementation details, see https://github.com/eli-quereli/ner-german-telegram/blob/main/src/notebooks/4.2_training/01_prepare_datasets/prepare_telegram_annotation_results.ipynb

($\sim 12\%$). Person names occur least frequently.

Table 6.1: Distribution of entities in Telegram corpus

	LOC	PER	ORG	DATE	TIME	ACTION	total
train	1.140	76	132	321	188	253	2.110
test	66	17	26	62	19	29	219
validation	73	17	18	40	9	23	180
total	1.279	110	176	423	216	305	2.509

6.2 The DFKI Smartdata Corpus

Compared to other NER datasets, the Telegram corpus is of very small size. The GermEval 2014 corpus for example consists of 31,300 sentences containing more than 41.000 named entities (cf. Benikova et al. 2014). However, creating a larger corpus would have required more resources for annotation. I therefore extended my dataset by adapting the Smartdata corpus (Schiersch et al. 2018) for my task.

The Smartdata corpus consists of 2,598 German-language manually annotated documents which were sampled from online resources covering three genres: newswire texts, Twitter, and traffic reports via RSS feeds from radio stations, police and railway companies. The dataset contains a total of 22,075 named entities and is available under a permissive CC-BY 4.0 license (cf. *ibid.*).

Even though still relatively small-sized compared to standard NER datasets, the corpus is useful for my purpose for two reasons: The corpus is designed for the recognition of events and therefore considers time and date mentions as named entities. Second, it was created from online documents and is thus more similar to the Telegram corpus than standard newswire corpora for NER.

However, the tag set of the corpus is not readily combinable with the Telegram corpus, since the Smartdata corpus is based on a (partially) fine-grained annotation scheme, tagging organization mentions as organization or organization-company, and location mentions as either location, location-city,

location-street, location-route, or location-stop. Furthermore, the dataset contains a number of entity types irrelevant for my task such as duration, distance, disaster type, or trigger events, while the entity type action from the Telegram corpus is missing. Last but not least, given the fact that the two corpora were annotated separately, annotation inconsistencies are likely to exist and negatively affect model training.

6.2.1 Label Mapping

Given the different tag sets, combining both datasets requires an adjustment of labels and annotations. I decided to use the tag set of the Telegram corpus as a reference, and adjust the Smartdata corpus accordingly.⁷ I mapped all irrelevant entity tags (e.g. duration, disaster type, trigger event) to the `O` tag, and most of the fine-grained entity tags to their coarse-grained equivalent (e.g. organization-company to organization), provided that definition of the fine-grained entity was considered to sufficiently match the entity definition from the Telegram corpus. Table A.2 in the appendix provides a detailed overview of the label mapping.

6.2.2 Adjusted Smartdata Corpus Statistics

The adjusted Smartdata corpus contains a total of 14,557 entities, distributed as shown in table 6.2. We can see that as in the Telegram corpus, locations represent the largest class ($\sim 47\%$), followed by organizations ($\sim 27\%$), and dates ($\sim 13\%$).

Table 6.2: Distribution of entities in Smartdata corpus after label mapping

	LOC	PER	ORG	DATE	TIME	ACTION	total
train	5.526	978	3.223	1.511	556	0	11.794
test	681	100	307	168	69	0	1.325
validation	688	127	387	167	69	0	1.438
total	6.895	1.205	3.917	1.846	694	0	14.557

⁷For implementation details, see https://github.com/eli-quere/ner-german-telegram/blob/main/src/notebooks/4.2_training/01_prepare_datasets.

6.3 The Final Corpus

Combining the Telegram corpus and the adjusted Smartdata corpus resulted in a total of 3,098 documents including 17,066 named entities with a class distribution of as shown in table 6.3.

Locations make up for almost half of all entities ($\sim 48\%$) in the combined corpus, followed by organizations ($\sim 24\%$), and date mentions ($\sim 13\%$). Less frequently mentioned are persons ($\sim 8\%$), and times ($\sim 5\%$). Action mentions make up for less than 2%, resulting from the absence of this entity type in the Smartdata corpus. The dataset is thus heavily imbalanced which is known to negatively affect training.

Table 6.3: Distribution of entities in the final corpus

	LOC	PER	ORG	DATE	TIME	ACTION	total
train	6.666	1.054	3.355	1.832	744	253	13.904
test	747	117	333	230	88	29	1.544
validation	761	144	405	207	78	23	1.618
total	8.174	1.315	4.093	2.269	910	305	17.066

6.4 Models

I fine-tuned five different pretrained transformer models from the three different encoder model architectures BERT, DistilBERT, and XLM-R. The models differ not only in terms of their architecture, but also regarding size, sources, and languages of the corpora used for pre-training.

6.4.1 BERT Models

BERT models are language models pretrained on the tasks of masked language modeling and next sentence prediction (cf. Devlin et al. 2018). I selected two German-language BERT models for fine-tuning:

dbmdz/bert-base-german-uncased

This model is a German-language model pretrained on 16GB of data taken from Wikipedia, but also large online corpora created via web crawlers. I

used the uncased version of the model, meaning that the model does not consider capitalization of words, since capitalization in user-generated text data is often unreliable (see section 2.6). The cased version of the model was evaluated for NER fine-tuning on the CoNLL 2003 and GermEval2014 datasets, achieving F1 scores of 84.52 and 86.89.⁸

Model card: <https://huggingface.co/dbmdz/bert-base-german-uncased>

deepset/gbert-base

This German-language BERT model was pretrained on 163.4GB of data mainly consisting of scraped online resources (cf. Chan, Schweter, and Möller 2020). Fine-tuning for NER was evaluated on the GermEval2014 dataset, with an F1 score of 87.98 (cf. *ibid.*).

Model card: <https://huggingface.co/deepset/gbert-base>

6.4.2 DistilBERT Model

DistilBERT models are distilled version of BERT models, which allow faster training and require less memory, while achieving accuracy close to regular BERT models (cf. Tunstall et al. 2022, p. 80).

distilbert-base-multilingual-cased

This model is a distilled version of a multilingual BERT model, pretrained on a masked language modeling task using a concatenation of Wikipedia in 104 languages. It is reported to be twice as fast as its BERT counterpart.⁹ The model is cased and thus considers capitalization. Evaluation results for NER fine-tuning are not provided.

Model card: <https://huggingface.co/distilbert-base-multilingual-cased>

⁸See <https://github.com/stefan-it/fine-tuned-berts-seq>, accessed on 2022/06/08.

⁹<https://huggingface.co/distilbert-base-multilingual-cased>, accessed on 2022/06/08.

6.4.3 XLM-Roberta Models

RoBERTa models modify the pretraining scheme of BERT models by dropping the task of next sentence prediction, and are trained longer and with more training data. This has shown to improve results compared to original BERT models (cf. Tunstall et al. 2022, p. 80). XLMs are cross-lingual language models. Their pretraining includes inputs from multiple languages, and extends the masked language modeling task to translation language modeling (cf. *ibid.*, p. 80). XLM-Roberta models combine these two approaches by performing multilingual pretraining with large amounts of training data (cf. *ibid.*, p. 80).

xlm-roberta-base

This model was pretrained on 2.5TB of webcrawled data covering more than 100 languages (cf. Conneau et al. 2019). It was evaluated for NER fine-tuning on the CoNLL 2002 and CoNLL 2003 datasets, achieving an F1-score of 84.60 for German when fine-tuned on each language set separately (cf. *ibid.*).

Model card: <https://huggingface.co/xlm-roberta-base>

cardiffnlp/twitter-xlm-roberta-base

The last model is an XLM-RoBERTa model pretrained on ~198M tweets in over thirty languages (cf. Barbieri, Anke, and Camacho-Collados 2021), and thus the only model pretrained on social media texts. The evaluation of the model in the reference paper focuses on Twitter-specific tasks like e.g. emoji prediction, irony, hate speech and offensive language detection, or sentiment analysis. Evaluation of NER fine-tuning is not provided (cf. *ibid.*).

Model card: <https://huggingface.co/cardiffnlp/twitter-xlm-roberta-base>

6.5 Experimental setup

In a first iteration, each model was fine-tuned on the Telegram corpus and the adjusted Smartdata corpus separately. The second iteration consisted of

fine-tuning each model on the combined corpus. Before I report my results, I will outline the necessary steps for training and evaluation. The procedure was the same for all models to ensure comparability of results.

6.6 Defining a Baseline

Defining an appropriate baseline for my task is difficult due to a various reasons: The evaluation results for German-language NER fine-tuning for the chosen models are not directly comparable, since the standard datasets CoNLL2003 and GermEval2014 were used. These datasets differ greatly from my training corpus in terms of size, genre, and number of classes.

The shared tasks on social media NER described in ?? are roughly comparable to my setup in terms of corpus size, genre, and number of classes. However, all systems used in them represent outdated approaches to NER. This fundamentally limits the comparability of their results with my findings.

I decided to focus on state of the art results. Hence I will use the F1 score of the `deepset/gbert-base` model fine-tuned on the GermEval 2014 corpus. My baseline is thus $F1=87.98$.

6.7 Fine-tuning an NER model

The following implementation steps for NER fine-tuning were adapted from Tunstall et al. 2022, pp. 87–122.¹⁰

6.7.1 Preprocessing: Tokenization and Encoding

Tokenizers for transformer models takes care of several pre-processing steps including normalization, pretokenization, subword tokenization, and post-processing (cf. *ibid.*, pp. 94–95). Transformer models rely on subword tokenization, combining character- und word-based tokenization by splitting up

¹⁰For implementation details, see https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/4.2_training/02_training.

words rarely occurring in the training data into smaller units and retaining frequent words as a whole. This allows to handle complex words and misspellings, but at the same time keep the input length at a processable size (cf. Tunstall et al. 2022, p. 33). During pretraining, different algorithms such as WordPiece, Byte-Pair Encoding (BPE), or SentencePiece are applied to the training data, allowing the model to learn which words should be split up further (cf. HuggingFace 2022d). Tokenizers for my experiments were loaded via the `from_pretrained()` method of the `AutoTokenizer` class.

Label alignment for tokenized inputs

While the tokenizer splits up the input strings into subwords, the named entity tags in the training corpus are usually assigned to entire words. Therefore we need to take care of aligning the tags with their corresponding subtokens.

Every word in the input is assigned a unique `word_id`. This `word_id` also assigned to all corresponding subtokens. We can therefore use this id for label alignment. Subword tokenization also needs to be considered for later evaluation, since the loss function of the model needs to know how to handle subtokens. My implementation follows Tunstall et al. and only considers the first subtoken of a word for calculating the loss. This can be implemented by assigning the value -100 as `label_id` to all subtokens which have the same `word_id` as their predecessor. This tells the cross entropy loss function to ignore these subtokens, hence only the model's prediction for the first subtoken of a word will be considered (cf. Tunstall et al. 2022, p. 104). I used `is_split_into_words=True` to tell the tokenizer that the input data is already tokenized. Shorter inputs were padded, longer inputs were truncated to the maximum length of the respective model.

After preprocessing and encoding, the dataset includes numerical `input_ids`, representing the input tokens, and numerical `labels`, representing the corresponding named entity tags, as shown in figure 6.2. The tokenizer has also returned an `attention_mask` for each input document, telling which token should be attended and which not (cf. HuggingFace 2022a). Furthermore, the tokenizer created `token_type_ids`. These are not relevant for my task,

but used for tasks in which an input contains two sequences, e.g. for question answering or tasks involving sentence pairs such as next sentence prediction. In these cases, the `token_type_ids` tell the model to which of the two sequences a token belongs (cf. HuggingFace 2022b). We now have encoded our textual input, consisting of tokens and labels, into a numerical input which can be processed by the model.

	input_ids	token_type_ids	attention_mask	labels
0	[102, 20739, 8497, 30914, 10412, 30925, 22171,...]	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...]	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...]	[-100, 0, -100, -100, -100, -100, -100, -100, ...]
1	[102, 17329, 14705, 18315, 17329, 19281, 30904,...]	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...]	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...]	[-100, 0, -100, -100, 5, -100, -100, -100, -100, ...]
2	[102, 718, 1412, 6556, 17077, 3859, 136, 8139,...]	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...]	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...]	[-100, 0, 3, -100, -100, -100, 0, 3, -100, 0, ...]
3	[102, 493, 566, 1483, 1073, 818, 3035, 853, 32...]	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...]	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...]	[-100, 1, 2, 2, 2, 0, 9, -100, -100, 10, 5, 0, ...]
4	[102, 15591, 17329, 7938, 2032, 305, 1055, 853...]	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...]	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...]	[-100, 0, 0, -100, -100, -100, -100, 0, 0, 0, ...]

Figure 6.2: Training data after tokenization and encoding

6.7.2 Loading and Configuring a Model

As a next step, we need to load the model and adjust its configuration to make it fit our dataset. We do this by loading the `AutoConfig` of the pretrained model and handing over the number of labels (`num_labels`) in our dataset as well as the mapping between numerical and textual labels, defined as (`id2label`) and (`label2id`). Since NER fine-tuning is about classification of single tokens for NER, we choose a model for token classification.

6.7.3 Defining Training Arguments

The following training arguments were defined for all models and datasets (some arguments for logging and saving the model are omitted):

```
from transformers import EarlyStoppingCallback

BATCH_SIZE = 2
NUM_EPOCHS = 10
WEIGHT_DECAY = 0.01
CALLBACK = EarlyStoppingCallback(early_stopping_patience=3)
LOAD_BEST_MODEL_AT_END = True,
METRIC_FOR_BEST_MODEL='eval_loss'
```

By `NUM_EPOCHS` we define how many times the model will work through the whole dataset. `BATCH_SIZE` defines the number of examples processed per device (e.g. a GPU) in each training step. A batch size of 2 is admittedly small, but was chosen to avoid running into memory errors during training.

`WEIGHT_DECAY` is used as one out of many regularization technique to prevent overfitting. I also defined a `CALLBACK` to implement early stopping during training. Callbacks can be used to customize the training behaviour of transformer models and take decisions during training. Here, the decision is to stop if the provided `METRIC_FOR_BEST_MODEL` deteriorates the third time in a row, defined via `early_stopping_patience=3`. I used the model's loss on the validation set as reference metric for early stopping and loading the best model at the end of the training process.

Loading a Data Collator

I also used a data collator which takes care of padding for labels and inputs. It should be noted however that padding was already carried out by the tokenizer. Using a data collator is an alternative to padding via the tokenizer, using both of them together is in fact redundant.

6.7.4 Defining Evaluation Metrics

For evaluation purposes, I defined the method `compute_metrics` which returns the F1 score, precision, recall, and accuracy per training epoch. Analogue to the process of label alignment in section 6.7.1, the model's predictions are aligned using the `label_id -100` for subtokens to be ignored by the

loss function.

6.7.5 Training

We can now initialize a `Trainer` and pass the loaded model, the training arguments, the tokenizer, the data collator, the callback, and the encoded data splits for training and validation to it, and then start the training.

The `Trainer`'s method `create_model_card()` was used as a convenient way to document the respective training arguments and evaluation results.

6.8 Results

The following section explores the results of fine-tuning the selected models on the two datasets separately, and on the final combined dataset. Recall the baseline is defined as $F1=87.98$.

Generally we can observe that for all experiments, training was stopped after a maximum of three epochs, in some cases already after one epoch, indicating that the validation loss starts to increase rather soon in all experiments. This is not surprising given the rather small size of the provided data sets, which easily leads overfitting.

The defined baseline could not be reached or outperformed in any of my experiments. Regarding the separate experimental setups, it can be observed that fine-tuning exclusively using the Telegram corpus overall yields the lowest results: None of the F1 scores increases 0.8 and are thus far from the baseline. The DistilBERT model achieves the lowest score with $F1=0.6481$, the German-language *deepset/gbert-base* model performs best with $F1=0.7482$.

Fine-tuning using only the adjusted Smartdata corpus yields the most promising results, as shown in table 6.5.

Overall, the models perform better with most F1-scores exceeding $F1=0.8$ and therefore getting closer to the baseline. An exception is the *dmbdz/bert-base-german-uncased* model achieving $F1=0.6156$. Once again, the *deepset/gbert-*

Table 6.4: Fine-tuning different models on Telegram corpus

Model	F1 score	Validation Loss	Epoch
distilbert-base-multilingual-cased	0.6481	0.2910	1
dbmdz/bert-base-german-uncased	0.7440	0.2285	2
deepset/gbert-base	0.7482	0.2524	2
xlm-roberta-base	0.7330	0.2459	3
cardiffnlp/twitter-xlm-roberta-base	0.7103	0.2648	3

base model performs best with F1=0.8535 which is only ~ 0.2 below the baseline, which is promising especially when considering that the baseline was achieved by fine-tuning on a much larger corpus.

Table 6.5: Fine-tuning different models on adjusted Smartdata corpus

Model	F1 score	Validation Loss	Epoch
distilbert-base-multilingual-cased	0.8203	0.1456	2
dbmdz/bert-base-german-uncased	0.6156	0.2560	3
deepset/gbert-base	0.8535	0.1114	3
xlm-roberta-base	0.8264	0.1445	1
cardiffnlp/twitter-xlm-roberta-base	0.8266	0.1344	3

The results for fine-tuning on the combined corpus are shown in table 6.6. We can observe that the scores improved compared to fine-tuning only on the Telegram corpus, which is plausible considering the larger size of provided data. However, we notice a decrease of scores compared to fine-tuning only on the adjusted Smartdata corpus. One explanation for this could be annotation inconsistencies between the two corpora, or the imbalanced class distribution, especially the low number of examples for the entity type action.

Table 6.6: Fine-tuning different models on combined corpus

Model	F1 score	Validation Loss	Epoch
distilbert-base-multilingual-cased	0.6634	0.2543	2
dbmdz/bert-base-german-uncased	0.7775	0.1751	1
deepset/gbert-base	0.7922	0.1751	2
xlm-roberta-base	0.7988	0.1842	3
cardiffnlp/twitter-xlm-roberta-base	0.7768	0.2061	3

Judging only by the F1-score, the best model on the combined corpus was the *xlm-roberta-base* model with F1=0.7988, followed closely by the *deepset/gbert-*


```
df.drop_duplicates()

return df
```

We can visualize the results by creating some simple line plots with the following code:

```
eval_df = get_training_history(trainer)

# plot train, eval loss
sns.set_style("whitegrid")
ax = sns.lineplot(data = eval_df[['training_loss', 'validation_loss']])
ax.set_xticks(range(len(eval_df)), labels=range(len(eval_df)))
plt.show()

# plot f1, precision, recall, accuracy
sns.set_style("whitegrid")
ax = sns.lineplot(data = eval_df[['f1', 'precision', 'recall', 'accuracy']])
ax.set_xticks(range(len(eval_df)), labels=range(len(eval_df)))
plt.show()
```

Figure 6.3 and 6.4 show evaluation metrics for the fine-tuned deepset/gbert-base model. We can make the following observations:

The loss on the training set intersects with the validation loss between epoch 1 and 2. After that, the training loss continues to decrease, while the validation loss remains relatively constant, and then increases at epoch 5. This can be interpreted as overfitting, as we have already observed when looking at the number of epochs in the results above.

In the second plot we see that accuracy is continuously high with around 0.95. This is, however, not very informative for an NER model, since we are dealing with a very imbalanced dataset (see section ??).

We can furthermore observe that recall for this model is in general higher than its precision. This means that the model tends to return less false

negatives than false positives. Its predictions are thus better in terms of completeness than in terms of relevance or validity.

```
from torch.nn.functional import cross_entropy

num_labels = 13

def forward_pass_with_label(batch):
    # Convert dict of lists to list of dicts suitable for
    # data collator
    features = [dict(zip(batch, t)) for t in zip(*batch.
                                                values())]

    # Pad inputs and labels and put all tensors on device
    batch = data_collator(features)
    input_ids = batch["input_ids"].to(device)
    attention_mask = batch["attention_mask"].to(device)
    labels = batch["labels"].to(device)
    with torch.no_grad():
        # Pass data through model
        output = trainer.model(input_ids, attention_mask)
        # Logit.size: [batch_size, sequence_length, classes]
        # Predict class with largest logit value on classes
        # axis
        predicted_label = torch.argmax(output.logits, axis=-1
                                       ).cpu().numpy()

    # Calculate loss per token after flattening batch
    # dimension with view
    loss = cross_entropy(output.logits.view(-1, num_labels),
```



Figure 6.3: Fine-tuning the deepset/gbert-base model: Training and validation loss

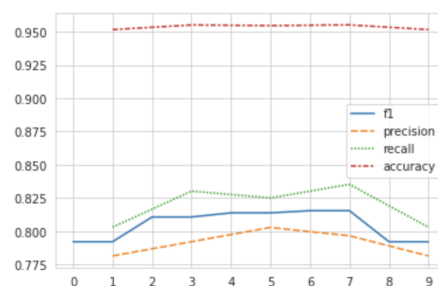


Figure 6.4: Fine-tuning the deepset/gbert-base model: F1, precision, recall, and accuracy


```

        labels.view(-1), reduction="none")
    # Unflatten batch dimension and convert to numpy array
    loss = loss.view(len(input_ids), -1).cpu().numpy()

    return {"loss":loss, "predicted_label": predicted_label}

valid_set_gbert = eval_encoded
valid_set_gbert = valid_set_gbert.map(forward_pass_with_label
                                     , batched=True, batch_size=32)
df = valid_set_gbert.to_pandas()

```

We can now further work with this dataframe to explore the loss on token and class level, and plot a confusion matrix.¹¹

Token-level errors

	0	1	2	3	4	5	6	7	8	9
input_tokens	.	-	Uhr	,	#	)	(	:	@	/
count	647	351	64	402	110	110	112	262	150	92
mean	0.17	0.27	0.75	0.11	0.4	0.38	0.3	0.11	0.18	0.24
sum	112.12	94.77	47.83	44.79	44.51	42.12	33.42	28.75	26.45	21.82

Figure 6.5: Tokens ranked by mean loss, German-language BERT model

Observations . token has highest total loss, not surprising since most common token in the list, and also part of named entities (count: 647), whitespace does not appear here (anders als bei roberta)

tokens: - # : Uhr () , / are rarer, but have relatively high mean loss -i appear often together with named entities, are even sometimes part of them, that's why model mixes them up; probably need to revise annotations / annotation guide

¹¹For implementation details see https://github.com/eli-quereli/ner-german-telegram/blob/main/src/notebooks/4.2_training/02_training/model_experiments_combined_datasets/gbert_smartdata_telegram.ipynb

Class-level erros

	0	1	2	3	4	5	6	7	8	9	10	11	12
labels	I- ACTION	B- ACTION	I-ORG	I-LOC	B- DATE	B- ORG	B- TIME	I- DATE	I-PER	I- TIME	B- PER	B-LOC	O
count	6	47	251	315	214	416	95	166	127	61	142	852	12862
mean	5.62	1.21	1.05	0.85	0.8	0.78	0.67	0.52	0.51	0.48	0.4	0.27	0.06
sum	33.71	56.65	263.46	267.29	171.45	324.4	63.18	86.67	64.89	29.38	57.11	228.98	821.87

Figure 6.6: Classes ranked by mean loss, German-language BERT model

- I-ACTION has highest loss, ua because very rare (only 6 occurences) -> need for more data
- same for B-ACTION (same with roberta-model, label is problematic, maybe drop or label more)
- I-ORG, I-LOC? - lowest loss: O
- lowest loss labels: B-LOC, B-PER, I-TIME, I-PER, I-DATE

Confusion matrix

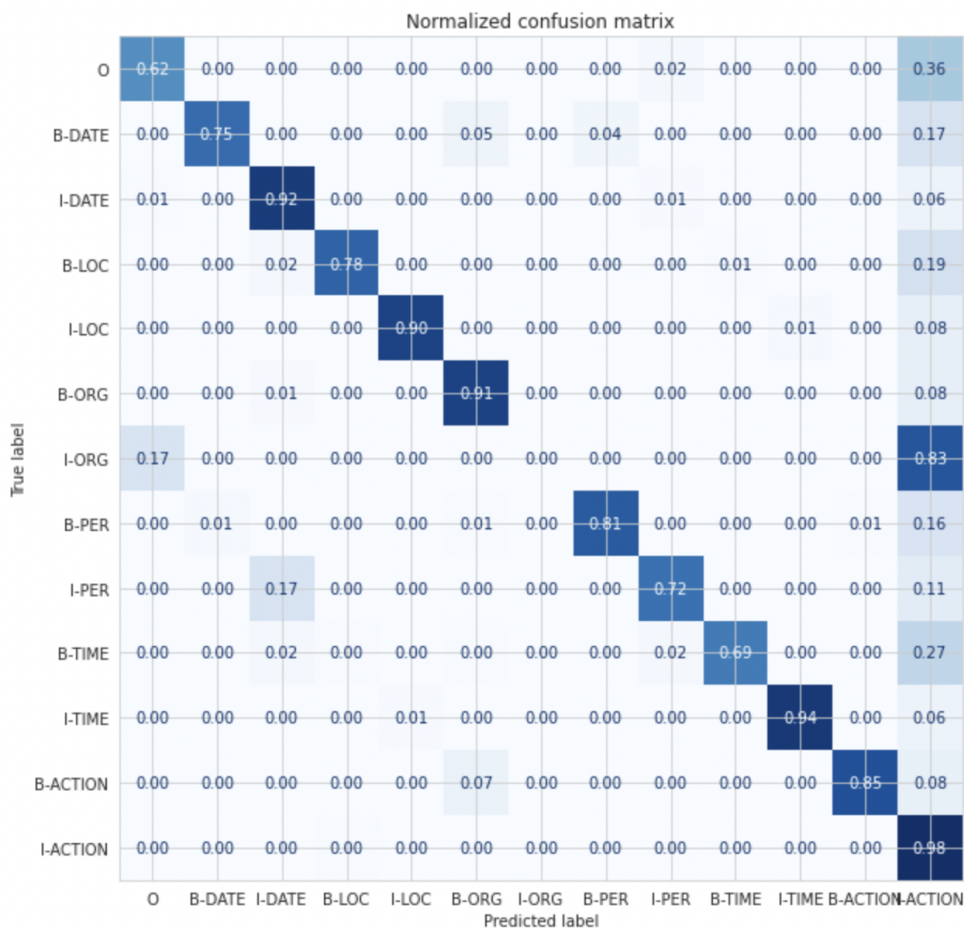


Figure 6.7: Confusion matrix, German-language BERT model

6.10 Summarization

Chapter 7

First Results: Applying the Custom NER Model to Telegram Messages

7.0.1 Example messages

Observations: model now detects time and date to some extent, did not recognize action Demo-Berlin, but unlike Flair model does not tag is as a location

makes errors, e.g. postcal code missing last digit, 0 as time entity, Kurpfalz as location instead of offene gesellschaft kurpfalz

Table 7.1 shows the results of the German-language Flair model and the custom fine-tuned model for the example texts:

7.0.2 Larger sample

CHAPTER 7. FIRST RESULTS: APPLYING THE CUSTOM NER MODEL TO TELEGRAM MESSAGES

Table 7.1: Detected entities for German-language example messages: Flair vs. Custom model

No.	Text	flair/ner-german	gbert-base/finetuned
1	Freiburg Aufzug mit Endkundgebung am Platz der alten Synagoge . Live Musik ! Details 19062021 Raus auf die Straße	Freiburg[LOC]	Freiburg[LOC]
2	GEHT DEMONSTRIEREN ! - Message von Elijah Tee an die Dresdner & anderen Demonstranten in Deutschland Morgen am 13.06. um 16.30 Uhr und bundesweit überall , wo es Mitdenker gibt . - Bilder und Videos hochladen - Diskussion Fahren oder Mitfahren	Elijah Tee[PER], Deutschland[LOC]	Elijah Tee[PER], Dresdner[LOC], Deutschland[LOC], 13.06.[DATE], 16.30 Uhr[TIME]
3	15831 Blankenfelde Mahlow Berliner Speckgürtel Ausfahrt Lichtenrade am 14.03.2021 14032021 Raus auf die Straße	Blankenfelde Mahlow[LOC], Lichtenrade[LOC]	1583[LOC], Blankenfelde[LOC], Mahlow[LOC], Berliner Speckgürtel[LOC], Ausfahrt[LOC], Lichtenrade[LOC], 14.03.2021[DATE]
4	2021 10 11 Sven Liebich fragt Ministerpräsident Haseloff nach dessen Verständnis von Versammlungsfreiheit - Sven Liebich 0:23 Raus auf die Straßen Übersicht / Overview	Sven Liebich[PER], Haseloff[PER], Sven Liebich[PER]	20211011[DATE], Sven Liebich[PER], Haseloff[PER], Sven Liebich[PER], 0[TIME]
5	FÜR ALLE DEMOKRATEN GEFAHR : Die Polizei blockiert den Stern ! Deswegen laufen die Menschen vom Ernst-Reuther-Platz nach Südosten zum Breitscheidplatz und dort weiter auf den Kudamm ! ! Schließt euch an : Neue Kantstraße direkt Richtung Osten auf den Breitscheidplatz ! Von der Straße des 17. Juni Richtung Süden zum Kudamm !	FU[ORG], Ernst-Reuther-Platz[LOC], Breitscheidplatz[LOC], Kudamm[LOC], Kantstraße[LOC], Breitscheidplatz[LOC], Kudamm[LOC]	Ernst-Reuther-Platz[LOC], Breitscheidplatz[LOC], Kudamm[LOC], Neue Kantstraße[LOC], Breitscheidplatz[LOC], 17.Juni[DATE], Kudamm[LOC]
6	Dr. Gunter Frank : Wie Moralismus dazu führt das man Feinde sieht und vernichten will — Antihygienista - Offene Gesellschaft Kurpfalz 15:04 Raus auf die Straßen Übersicht / Overview	Gunter Frank[PER], Antihygienista[ORG], Offene Gesellschaft Kurpfalz[ORG]	Gunter Frank[PER], Antihygienista-Offene Gesellschaft[ORG], Kurpfalz[LOC], 15:04[TIME]
7	Einfach nur Krank solche Politiker ! Was sollen wir mit ihm machen wenn unsere Kinder anhand dieser Impfung nen schaden erleiden ? Wer ist dann dafür verantwortlich , wen schmeißen wir dann in den Knast und nehmen sein ganzes Geld weg , sollen wir das mit dir machen ? Raus auf die Straße	<i>no entities detected</i>	<i>no entities detected</i>
8	27.11.2021 Frankfurt Durchsagen von gemischt mit Musik . Super Idee Für mehr Berichte folgen auf ... Raus auf die Straßen Übersicht / Overview	Frankfurt[LOC]	27.11.2021[DATE], Frankfurt[LOC]
9	Demo-Berlin : Der Mann mit dem Schild wurde soeben von der Polizei festgenommen . Circa 7 Einsatzwagen sind vor Ort und umzäunen den Park großräumig . Am Humboldthain gibt es nichts mehr zu gewinnen . # b2908 Raus auf die Straßen	Demo-Berlin[LOC], Humboldthain[LOC]	Berlin[LOC], Humboldthain[LOC]
10	Anthony Fauci : Definition von “ vollständig geimpft ” könnte geändert werden — Epoch Times Nachrichten 6:59 Raus auf die Straßen Übersicht / Overview	Anthony Fauci[PER], Epoch Times Nachrichten[ORG]	Anthony Fauci[PER], Epoch Times Nachrichten[ORG], 6:59[TIME]

Chapter 8

Conclusion

8.1 Summarization

8.2 Discussion

more data

improved annotation scheme (zb () / nicht taggen, # splitten), wie umgehen mit hashtags für Demos zB #b2908

Preventing the special characters in hashtags and user mentions from being split up was intended to serve for creating a realistic social media corpus and allow to include these special characters into the labels assigned during the annotation process. However, evaluation of training results indicate that it might indeed make sense to split up these special characters and simply treat them as separate tokens which are not treated as part of the tagged named entities.

Mixed language messages: It could also be argued that they should be included in the final training data, since they represent realistic social media data and NER systems should be able to handle such kinds of texts as well.

Smartdata: no NE action, This results in an underrepresentation of this

specific class when merging both datasets and might most probably cause problems during training, since there will not be enough examples for the model to learn, and mentions of actions occurring in the Smartdata corpus do not have a corresponding tag.

entity disambiguation

relationship tagging

future work -i main adaptation on large telegram corpus

Glossary

Information Extraction The task of extracting information from unstructured or semi-structured machine-readable data, e.g. texts.. 6

Named Entity Recognition The task of extracting named entities from text.. 6

Natural Language Processing Natural Language Processing combines approaches from computer science, linguistics, and artificial intelligence. The main focus is to build systems and computational techniques for the automated processing and analysis of human language.. 6

Acronyms

ACE Automatic Content Extraction. 6

BiLSTM Bidirectional Long Short Term Memory Network. 27

CoNLL Conference on Computational Natural Language Learning. 15, 18, 27, 30, 41–43

CRF Conditional Random Field. 27

DARPA Defense Advanced Research Projects Agency. 6

IE Information Extraction. 6, *Glossary*: Information Extraction

MUC Message Understanding Conference. 6

NER Named Entity Recognition. 6–13, 15–19, 25–28, 30, 31, 34, 37, 38, 41–43, 45, *Glossary*: Named Entity Recognition

NLP Natural Language Processing. 6, 8, 11, 12, 18, 19, 26, 27, *Glossary*: Natural Language Processing

OOV out of vocabulary. 13

TAC KBP Text Analysis Conference Knowledge Base Population. 6

W-NUT Workshop on Noisy User-generated Text. 17, 18, 37

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Appendix A

Additional Material

Table A.1: Overview of entity types of the English spaCy model

Entity type	Definition
CARDINAL	Numerals that do not fall under another type.
DATE	Absolute or relative dates or periods.
EVENT	Named hurricanes, battles, wars, sports events, etc.
FAC	Buildings, airports, highways, bridges, etc.
GPE	Countries, cities, states.
LANGUAGE	Any named language.
LAW	Named documents made into laws.
LOC	Non-GPE locations, mountain ranges, bodies of water.
MONEY	Monetary values, including unit.
NORP	Nationalities or religious or political groups.
ORDINAL	"first", "second", etc.
ORG	Companies, agencies, institutions, etc.
PERCENT	Percentage, including "%".
PERSON	People, including fictional.
PRODUCT	Objects, vehicles, foods, etc. (not services).
QUANTITY	Measurements, as of weight or distance.
TIME	Times smaller than a day
WORK_OF_ART	Titles of books, songs, etc.

Table A.2: Mapping of label schemas

Telegram Label	Original Smartdata Label
B-DATE	B-DATE
I-DATE	I-DATE
B-LOC	B-LOCATION, B-LOCATION_CITY, B-LOCATION_STOP, B-LOCATION_STREET
I-LOC	I-LOCATION, I-LOCATION_CITY, I-LOCATION_STOP, I-LOCATION_STREET
B-ORG	B-ORGANIZATION, B-ORGANIZATION_COMPANY
I-ORG	I-ORGANIZATION, I-ORGANIZATION_COMPANY
B-PER	B-PERSON
I-PER	I-PERSON
B-TIME	B-TIME
I-TIME	I-TIME
O	B-DISASTER_TYPE, I-DISASTER_TYPE, B-DISTANCE, I-DISTANCE, B-DURATION, I-DURATION, B-LOCATION_ROUTE, I-LOCATION_ROUTE, B-NUMBER, I-NUMBER, B-ORG_POSITION, I-ORG_POSITION, B-TRIGGER, I-TRIGGER

Annotation Guidelines

The following annotation guidelines are a combined adaptation of Benikova, Biemann, and Reznicek 2014, LDC 2005, Chinchor 1997, and Schön et al. 2018.

Named Entity - Basic Definition

Names are denominators for a single unique object in the real world. Only full noun phrases are considered as named entities. Pronouns and all other phrases can be ignored. Phrases only partly containing names (e.g. ‘deutschlandweit’), or adjectives referring to entities (‘euklidisch’) can be ignored.

Only one entity type per token should be annotated. In case of multiple options (e.g. ‘Charité’ could be referred to as organization or as location, same for ‘Landtag’ or ‘Bundestag’), the context of usage should be considered for the final classification.

Named entities can consist of more than one token. The initial token of an entity should be tagged with the B-label of the corresponding entity type. The following tokens belonging to the same entity should be labelled with the I-label of the same type. For

Combined named entities including different entity types (e.g. ‘Wien-Demo’) should be tagged separately if possible (Wien[LOC]-Demo[ACTION]), or else tagged as the dominant entity type (here: [ACTION]) to avoid nested entities.

Whitespace should not be labelled as part of an entity. Token separators such as /, - or . should be included if they are inside an entity sequence, and excluded if they separate two different entities, or an entity and a non-entity token. Hashtags should be included in the labeled span.

Dates and times can be represented by numbers (31.10.2019, 16:00), and strings ('Morgen', 'Uhr').

Emoticons, user mentions and (parts of) urls should not be tagged as named entities.

Definition of Different Entity Types

For annotation, the the 6 entity types PERSON, LOCATION, ORGANIZATION, DATE, TIME, and ACTION are used. The label set consists of B-tag and I-tag for each entity type, resulting in a total of 12 labels.

Table A.3 shows the core definitions and examples for the annotated entity types.

Specifications

Handling Mixed-language Messages

Skip mixed-language in which the actual message is non-German language.

Examples:

- *We are angry and pissed off - Raus auf die Straße Demotermine!*
- *Brilliant numbers in Carlisle ! Raus auf die Straßen @Demotermine*
- *MASSIMA DIFFUSIONE <http://t.me/stopdittatura/1930> Raus auf die Straßen @Demotermine*

Table A.3: Definition of entity types for annotation guide

Label	Description	Examples
PER	Named, single individual, limited to humans. Fictional persons can be included, as long as they're referred to by name.	Karl Lauterbach, Merkel, Harry Potter, Kretschmer, Karl Hilz, Markus
LOC	Names of politically or geographically defined locations, including countries, provinces, regions, cities, districts, addresses including streets, squares, and postal codes, (named) buildings or other man-made structures, shopping centers, parking lots, or restaurants, cardinal points. No metaphorical locations.	Deutschland, Frankreich, #BW, Berlin, Bundestag, Restaurant Walliserkanne, Straße des 17. Juni, Neptunbrunnen, Alex, Nollendorfplatz, Norden, Süden, Hellersdorf
ORG	A named corporate, governmental, or other organizational entity: Organization entities can be corporations, agencies, institutions, newspapers, tv/radio channels, other media platforms, political parties, non-governmental organizations, and political groups with a formal or informal organizational structure.	ARD, Polizei Berlin, Youtube, Telegram, Amadeu Antonio Stiftung, Querdenken711, Antifa, Bundesregierung, Gesundheitsamt, Französische Polizei, Facebook, Google, Bill Gates Foundation, KSV LiLi, FC St. Pauli, SGD-Ultras, Bündnis21 Hessen, Polizei, Polizeipräsidium Pforzheim
DATE	Absolute or relative date expressions	23.09.2022, 20210930, 3108, Morgen, Heute, 2019, #Sa2011
TIME	Specific, absolute time expressions referring to times of day, and time codes.	13:10, 13:12 Uhr, 16 Uhr, 6:49
ACTION	A specific instance of an action or event undertaken to express political beliefs, e.g. demonstrations, vigils, motorcades, walks in public.	Kundgebung, Demo, Demonstration, Spaziergang, Aufzug, Aufzug RuffderTrommeln

Handling Incomprehensible Messages

Skip incomprehensible messages.

Example: *“Sterne-Restaurant Walliserkanne . Naudreck . Bersets faule Tricks . Uniarzt panikfrei . Herzmuskel . Neuer Beitrag #Walliserkanne Raus auf die Straßen Demotermine! Übersicht / Overview”*

Handling Short Messages

Skip short messages which do not include at least two named entities.

Examples:

- *“AnwaelteFuerAufklaerung Demotermine”*
- *“Egal wo ihr diesen , oder die kommenden Samstage seid ! RheinCandleLight Demotermine”*

Handling Punctuation

Punctutaion should only be included in the tag if they appear in the middle of sequence representing an entity.

Example: *Donnerstag[B-DATE] ,[I-DATE] 17.04.2020[I-DATE]*

Hashtags

If a hashtag is part of a named entity, it should be included in the tag.

Examples:

- #Berlin[B-ORG]
- #BW[B-ORG]

- #CH[B-ORG]
- #Hardy[B-PER] Groeneveld[I-PER]
- #Mutigmacher[B-ORG]

If a hashtag is part of a sequence combining two named entities, it should be included in the tag for the first entity.

Example: #be[B-LOC]2908[B-DATE]

Detailed Annotation Instructions for Entity Types

PERSON

Person names can consist of given name and last name, or only first / only last name. Titles (e.g. Dr.) and forms of address (Herr, Frau) should not be annotated.

Examples:

- *Scholz*[B-PER]
- *Carolin*[B-PER] *Matthie*[I-PER]
- *Merkel*[B-PER]
- *Anselm*[B-PER] *Lenz*[I-PER]
- *Karl*[B-PER]

LOCATION

Do not label derived location words (e.g. *Kreuzberger Nächte* as location).

If a sequence contains a number of tokens representing a location entity (e.g. an address), use the B-tag for the initial token and the I-tag for all succeeding tokens belonging to the same entity, including special characters

such as , - . or /. The sequence should be long enough to create a representation of an entity which can be understood without the succeeding tokens.

Examples:

- *Straße[B-LOC] des[I-LOC] 17.[I-LOC] Juni[I-LOC]*
- *Bad[B-LOC] Oldesloe[I-LOC]*
- *Elbe[B-LOC] Elster[I-LOC]*
- *Bremen[B-LOC]-Vegesack[I-LOC]*
- *Lutherstadt[B-LOC] Wittenberg[I-LOC]*
- *Landeplatz[B-LOC] Griebkirchen[I-LOC]*
- *Google[B-LOC] center[I-LOC]*
- *Stuttgarter[B-LOC] Landtag[I-LOC]*
- NOTE: *Berlin[B-LOC] Neptunbrunnen[B-LOC]*

If the sequence contains a specification of the location, include the specification using the I-tags.

Examples:

- *Königstraße[B-LOC] Richtung[I-LOC] HBF[I-LOC]*
- *Bernauer[B-LOC] Str[I-LOC] Richtung[I-LOC] Süden[I-LOC]*
- *Marktoberdorf[B-LOC]-Kaufbeuren[I-LOC]-Neugablonz[I-LOC]*
- *Bremen[B-LOC]-Vegesack[I-LOC]*
- *D-67454[B-LOC] Haßloch[I-LOC]*

ORGANIZATION

If a sequence contains a number of tokens representing an organization entity, use the B-tag for the initial token and the I-tag for all succeeding tokens

belonging to the same entity, including special characters such as , - . or /. The sequence should be long enough to create a representation of an entity which can be understood without the succeeding tokens.

Examples:

- *Bundesregierung*[B-ORG]
- *Gesundheitsamt*[B-ORG]
- *Französische*[B-ORG] *Polizei*[I-ORG]
- *Google*[B-ORG]
- *Bill*[B-ORG] *Gates*[I-ORG] *Foundation*[I-ORG]
- *KSV*[B-ORG] *LiLi*[I-ORG]
- *FC*[B-ORG] *St.*[I-ORG] *Pauli*[I-ORG]
- *Bündnis21*[B-ORG] *Hessen*[I-ORG]
- *Polizeipräsidium*[B-ORG] *Pforzheim*[I-ORG]

TIME

Time expressions can include specific times of day and time codes (usually appearing after URLs. Sequences denoting a time span / duration should be annotated with the B-tag for the initial token and I-tags for the succeeding tokens.

Examples.

- *15:00*[B-TIME]
- *17:00*[B-TIME] *Uhr*[I-TIME]
- *www.example.com*[O] *00:30*[B-TIME]
- *Mittag*[B-DATE]

- *15:30[B-TIME] -[I-TIME] 19:30[I-TIME]*

DATE

Date expressions can be numeric values or strings. Sequences denoting a date spanshould be annotated with the B-tag for the initial token and I-tags for the succeeding tokens.

Examples:

- *2019[B-DATE]*
- *14.12.21[B-DATE]*
- *2.7.21[B-DATE]*
- *31. [B-DATE] JULI[I-DATE]*
- *01122020[B-DATE]*
- *Pfingstmontag[B-DATE]*
- *HEUTE[B-DATE]*
- *3108[B-DATE]*
- *201905[B-DATE]*
- *2019/03/05[B-DATE]*
- *28. [B-DATE]+ [I-DATE] 29.8. [I-DATE]*
- *17042021[B-DATE] -[I-DATE] 21042021[I-DATE]*

ACTION

Actions represent unique real-world events or actions. They must not necessarily be named, but be identifiable as unique object in the context in which they are mentioned. Also, the type of action should be expressed.

Examples:

- *Angriff*[B-ACTION]
- *Demo-B*
- *Autokorso*[B-ACTION]
- *Offenes*[B-ACTION] *Treffen*[I-ACTION]
- *Aufzug*[B-ACTION] *RufderTrommeln*[I-ACTION]
- *Zug*[B-ACTION]
- *Plakate*[B-ACTION]
- *Infostände*[B-ACTION]
- *Informationstouren*[B-ACTION]
- *Anti*[B-ACTION]-*Corona*[I-ACTION]-*Proteste*[I-ACTION]
- *Protestzug*[B-ACTION]
- *Spaziergänge*[B-ACTION]
- *Querdenker*[B-ORG]-*Demo*[B-ACTION]
- *Korso*[B-ACTION] *Nord*[I-ACTION]
- *Gedenken*[B-ACTION]
- *Aktionswoche*[B-ACTION]
- *Wahlkampf*[B-ACTION]
- *Live-Stream*[B-ACTION]

Statutory Declaration

I herewith formally declare that I have written the submitted thesis independently. I did not use any outside support except for the quoted literature and other sources mentioned in the paper.

I clearly marked and separately listed all of the literature and all of the other sources which I employed when producing this academic work, either literally or in content.

I am aware that the violation of this regulation will lead to the failure of the thesis.

.....	
Name	Signature

.....	
Matriculation Number	Place and Date